吉林大学
JILIN UNIVERSITY

Academic lecture of Organic Chemistry TII
College of Chemistry, Jilin University



Metal hydride hydrogen atom transfer reactions

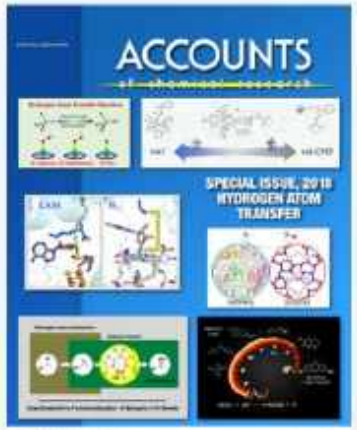
Shuo Zhang

2022.04.08

What is the meaning of Metal hydride hydrogen atom transfer reactions

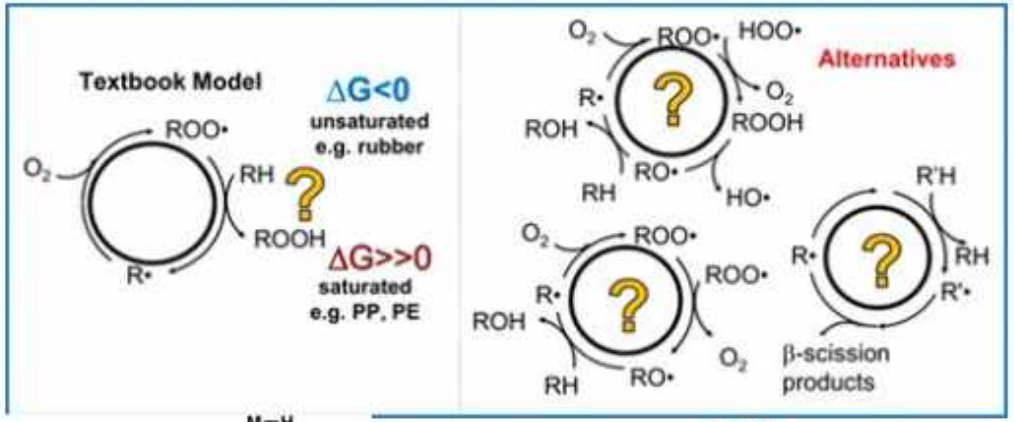
■ The growing application of HAT in a number of fields

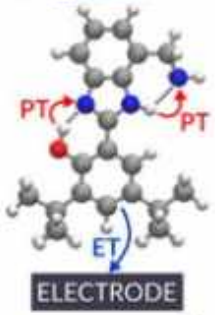
Hydrogen Atom Transfer

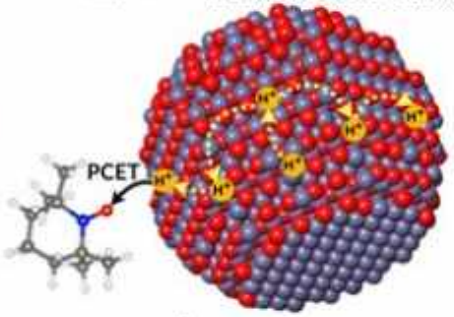


The study of hydrogen atom transfer (HAT) reactions has witnessed tremendous advancement in recent years from both experimental and theoretical points of view, in parallel to the growing application of these processes in a number of fields. This Special Issue, guest-edited by Miquel Costas and Massimo Bietti, presents an overview on the most important aspects and most recent developments of HAT processes, with the final goal of providing a reference tool for both the practitioner and the newcomer to the field.

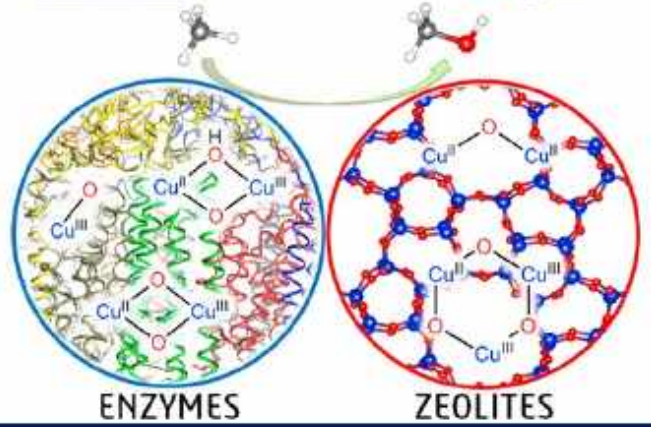
Hydrogen Atom Transfer Special Issue (acs.org)



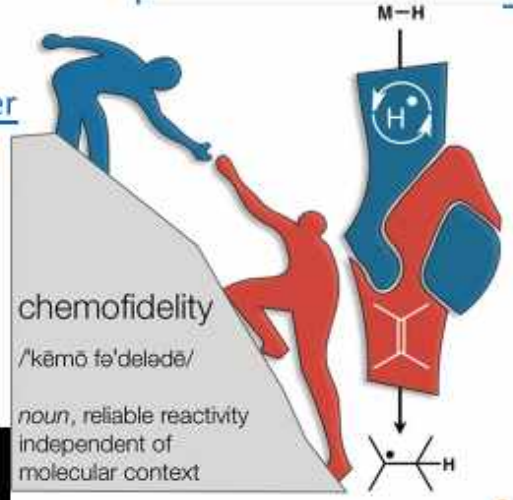




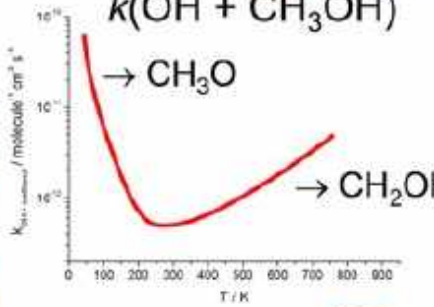


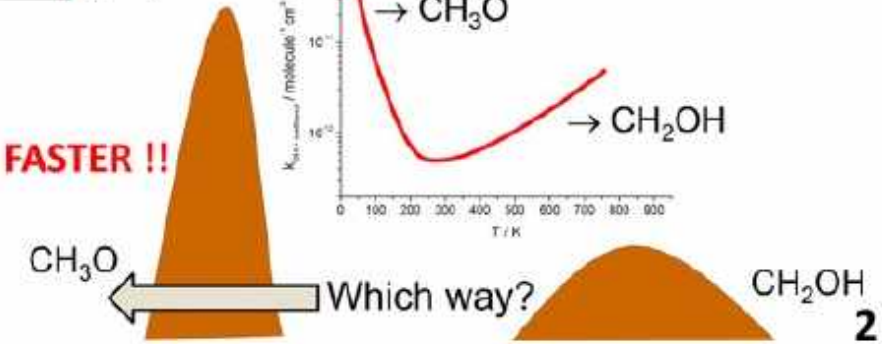






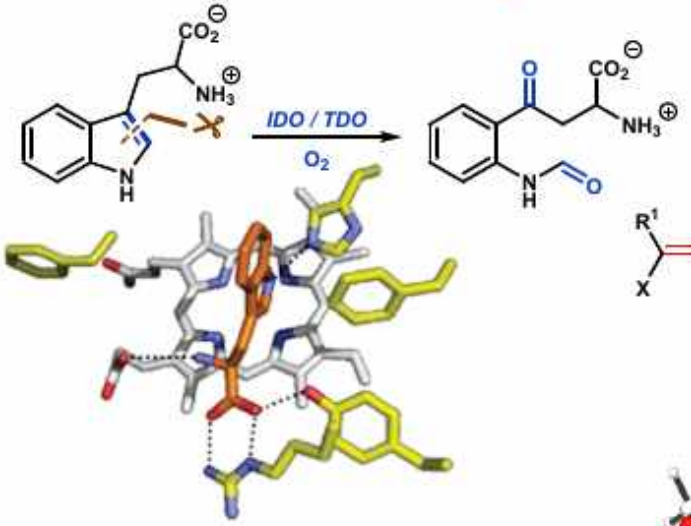




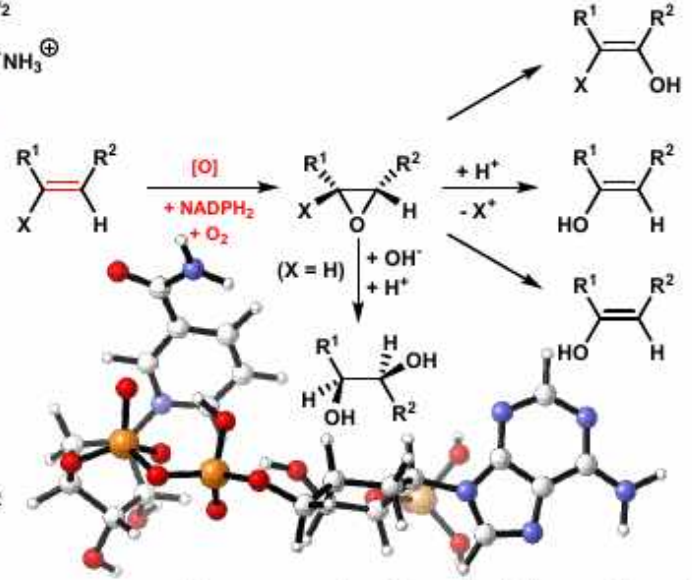


What is the meaning of Metal hydride hydrogen atom transfer reactions

■ The History and Beginning

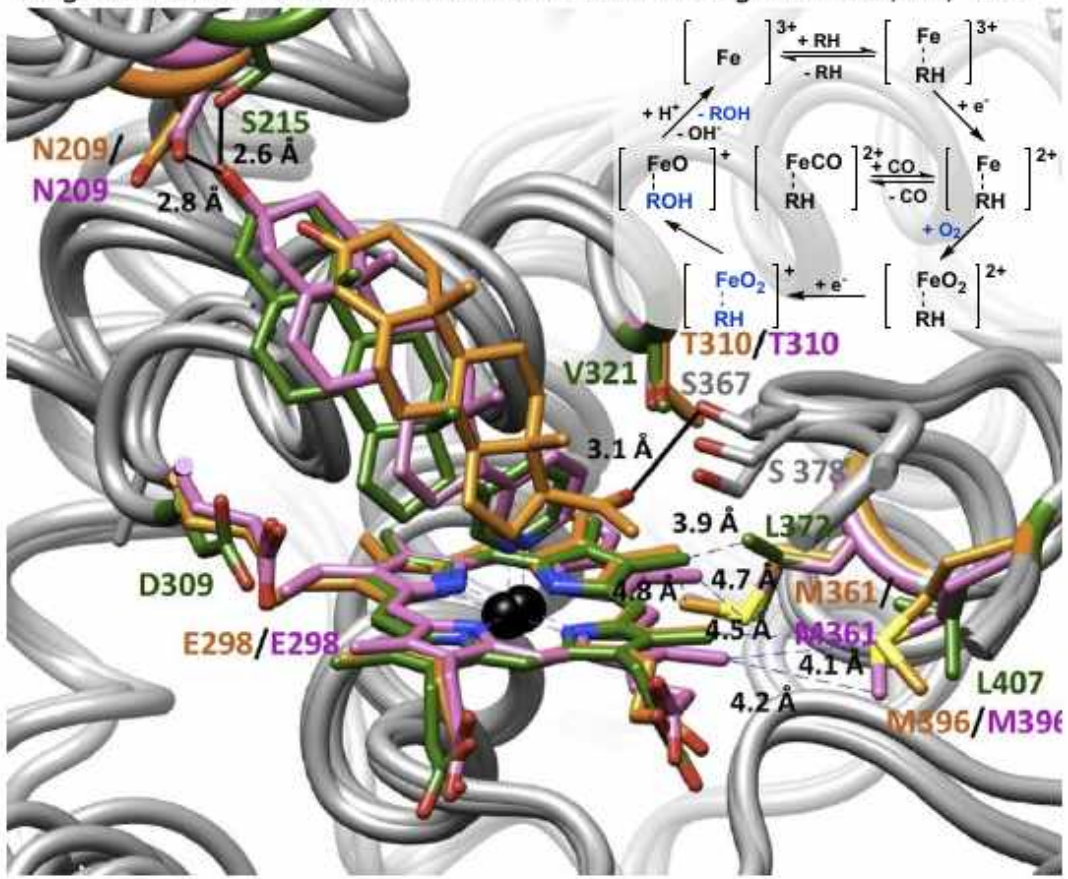


Exquisite control over the reactivity with O_2
Biochemistry **2011**, *50*, 2717.

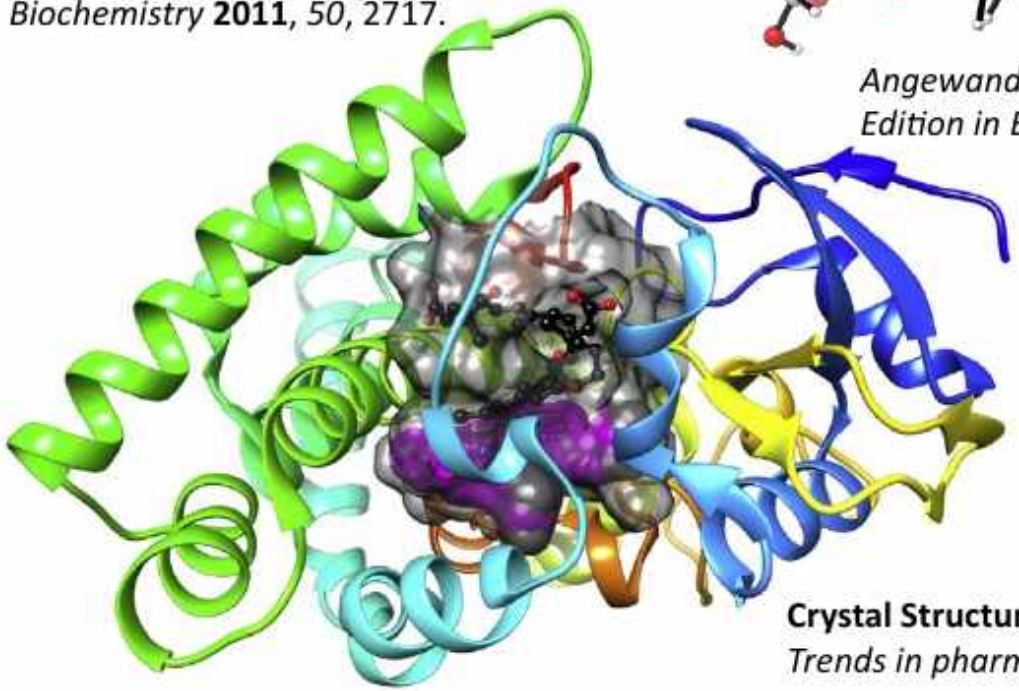


Angewandte Chemie International Edition in English **1972**, *11*, 701.

The Course of the Cytochrome P-450-Catalyzed Monooxygenation
Angewandte Chemie International Edition in English **1972**, *11*, 701.



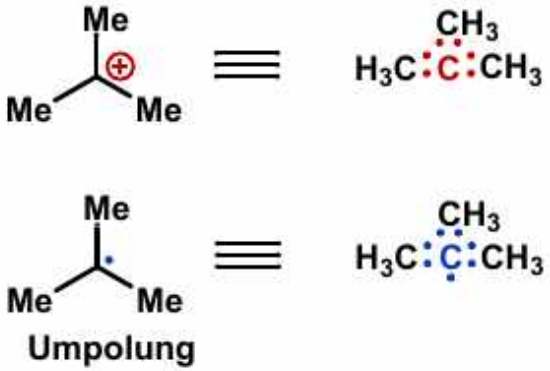
Similar Tertiary Structure Folds of Human P450
Trends in pharmacological sciences **2016**, *37*, 625.



Crystal Structure and Active Site Volume of Bacterial P450rev
Trends in pharmacological sciences **2016**, *37*, 625.

What is the meaning of Metal hydride hydrogen atom transfer reactions

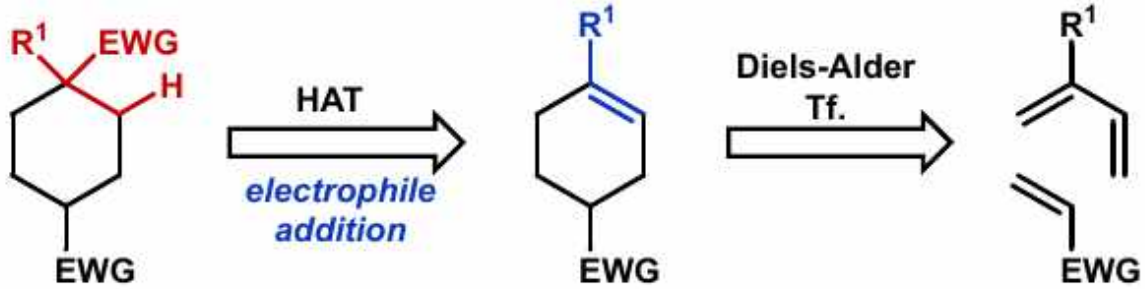
- The importance derive mainly from two attributes

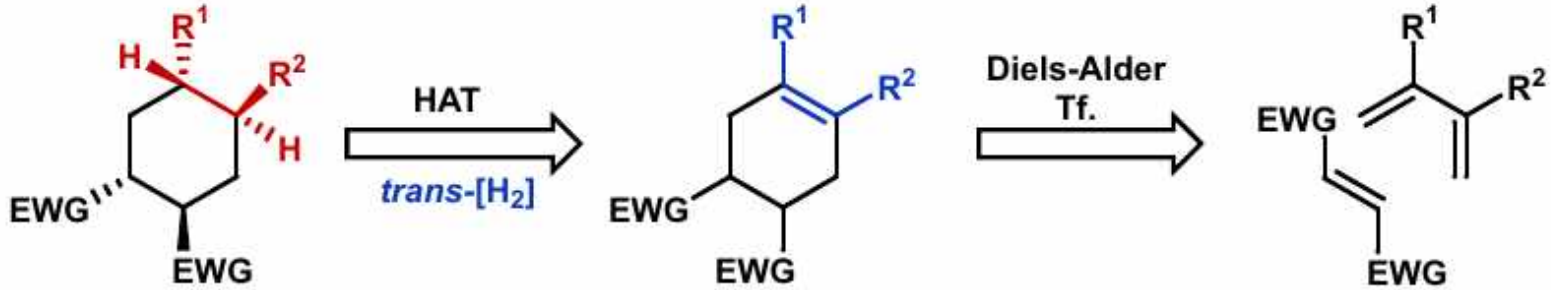


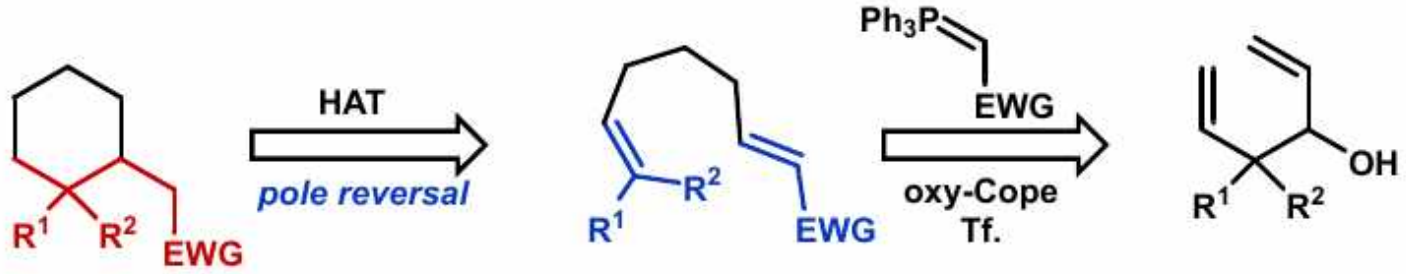




Markovnikov hydrofunctionalization of alkenes



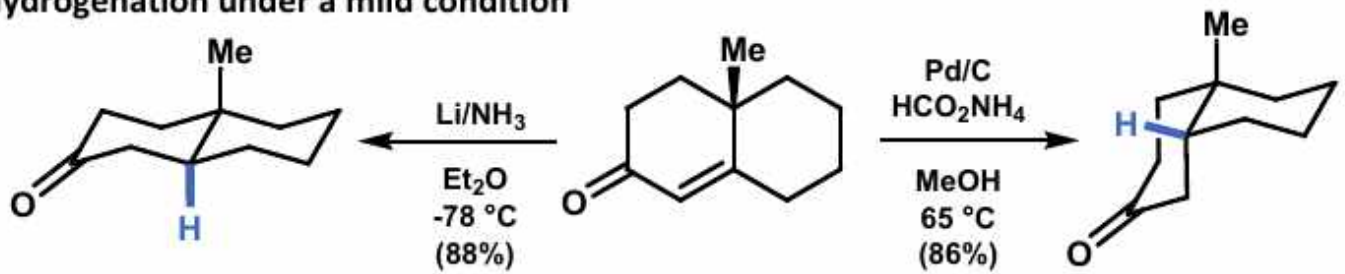


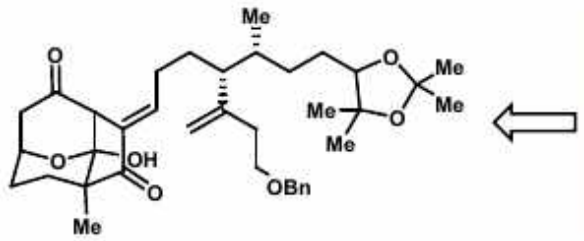


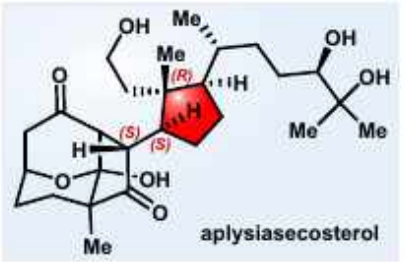
What is the meaning of Metal hydride hydrogen atom transfer reactions

■ The importance derive mainly from two attributes

Hydrogenation under a mild condition

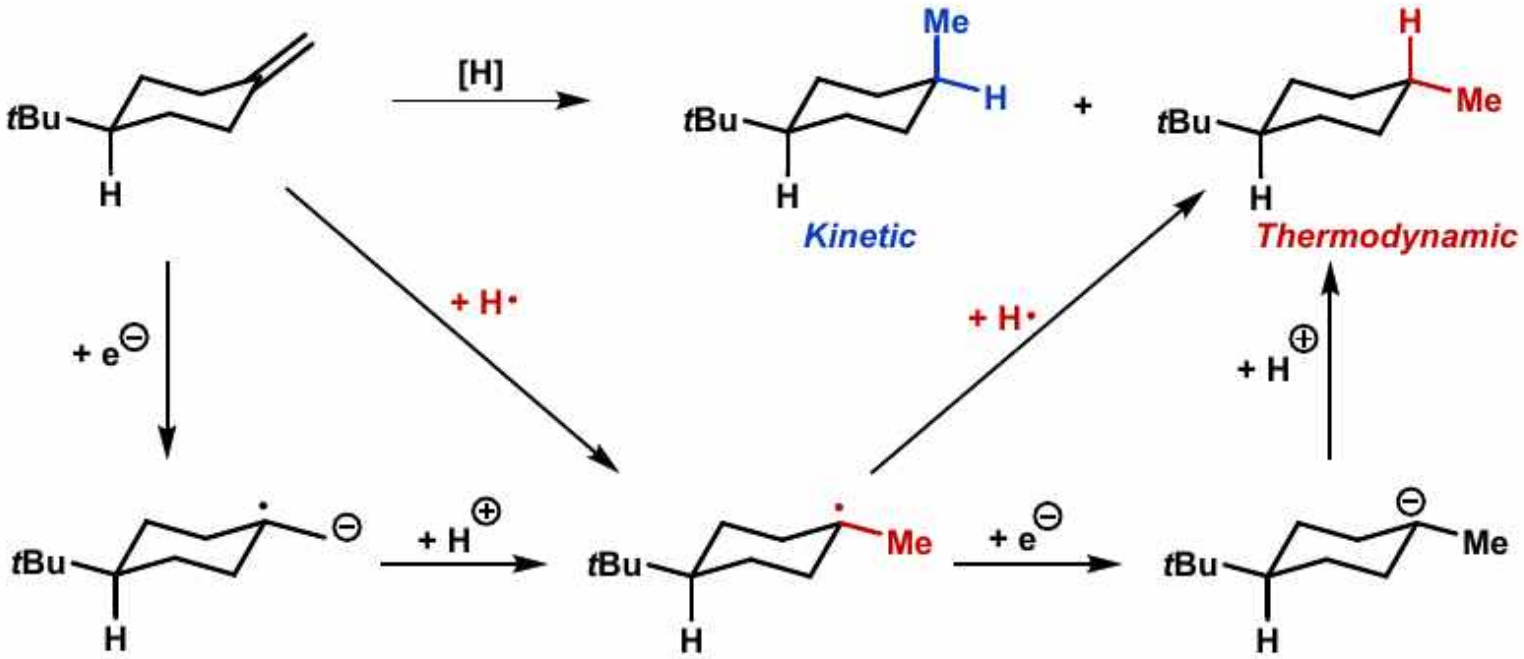




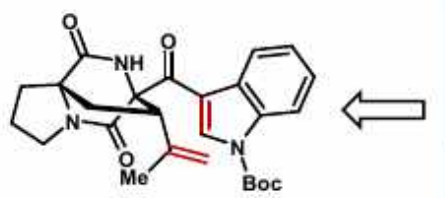


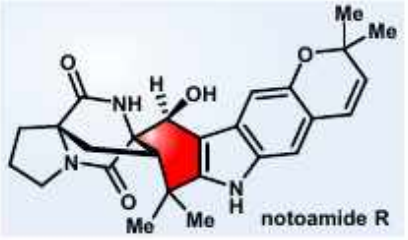
Accounts of chemical research **2018**, *51*, 2628.

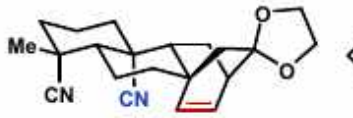
Form a thermodynamically stable intermediate through a special way

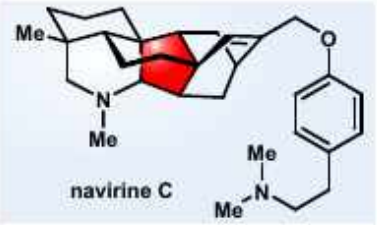


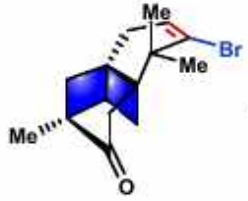
Chemical reviews **2016**, *116*, 8912.

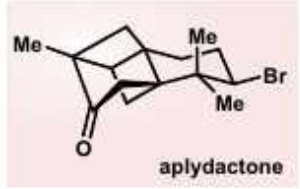






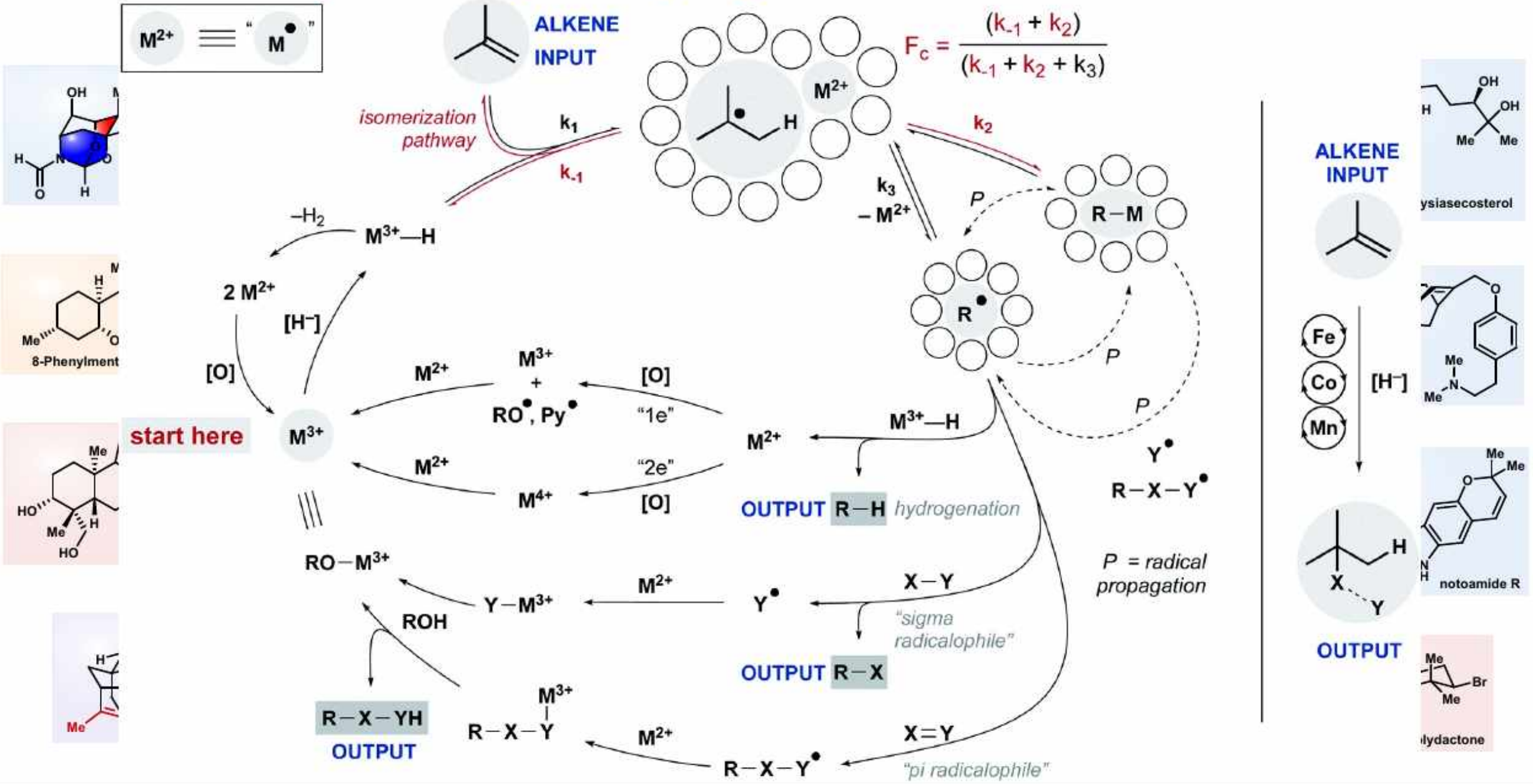






Accounts of chemical research **2018**, *51*, 2628.

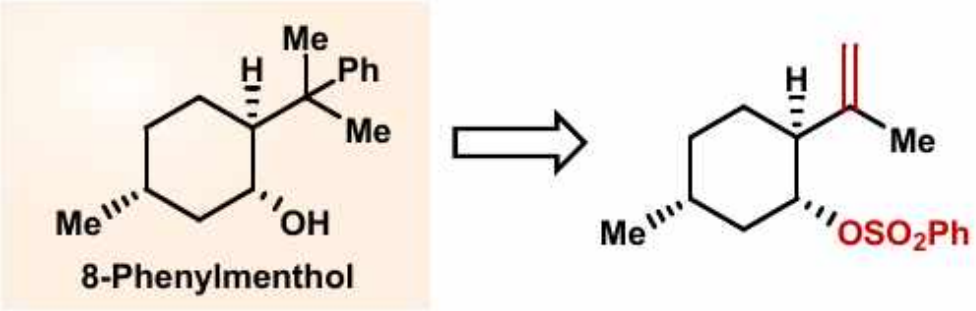
What is the meaning of Metal hydride hydrogen atom transfer reactions

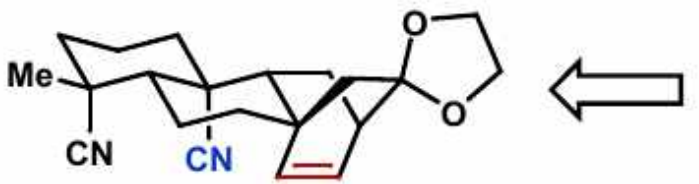


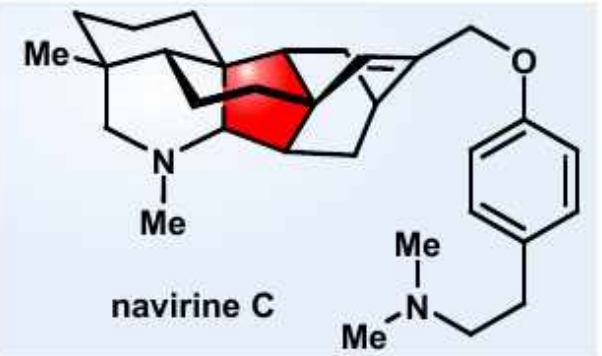
What is the meaning of Metal hydride hydrogen atom transfer reactions

- Chemoselectivity, Chemofidelity and Application in Cascade reaction

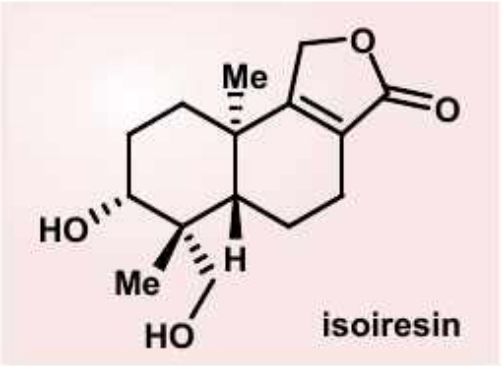
Synthesis of 8-Arylmenthol Derivatives via a Smiles-Truce rearrangement

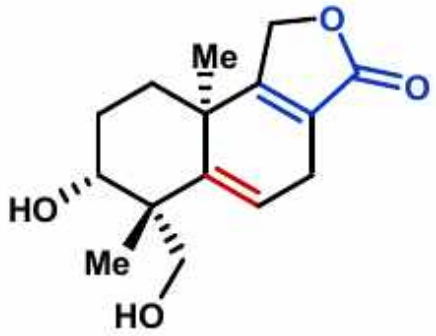


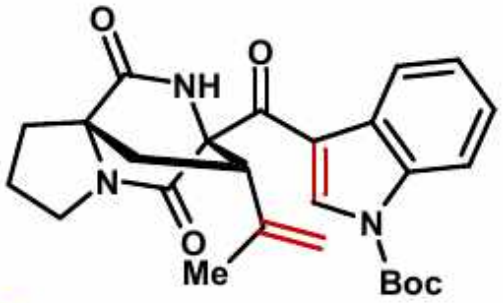


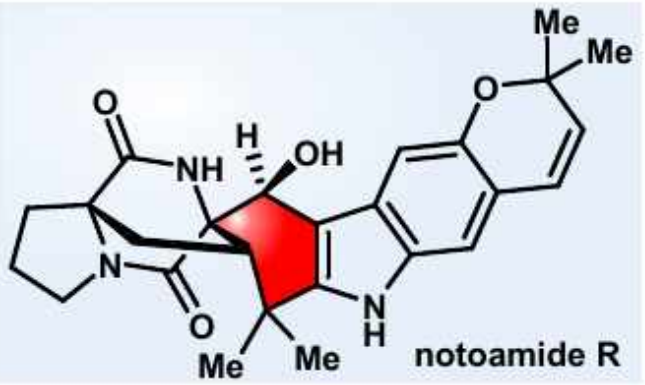


The Wonderful Chemoselectivity







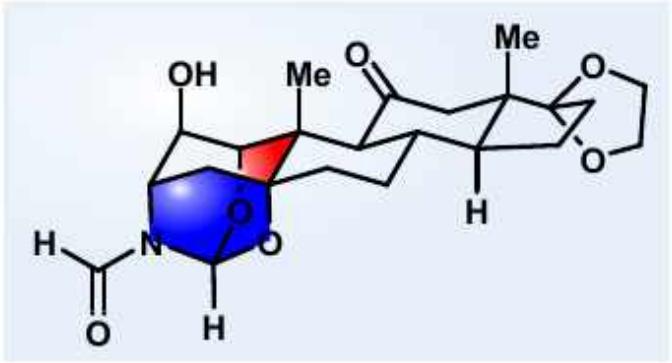


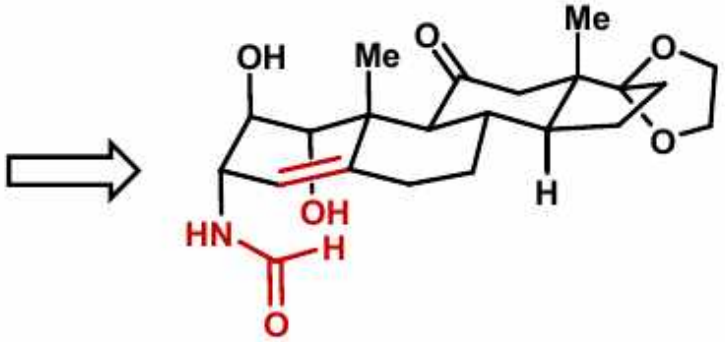
Construction of molecular skeleton using cascade reaction and the great Chemoselectivity

Green, S. A.; Crossley, S. W.; Matos, J. L.; Vásquez-Céspedes, S.; Shevick, S. L.; Shenvi, R. A.
Accounts of chemical research **2018**, *51*, 2628.

What is the meaning of Metal hydride hydrogen atom transfer reactions

- Chemoselectivity, Chemofidelity, Mild Condition and Thermodynamic selectivity

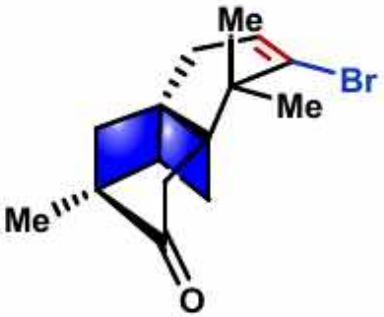


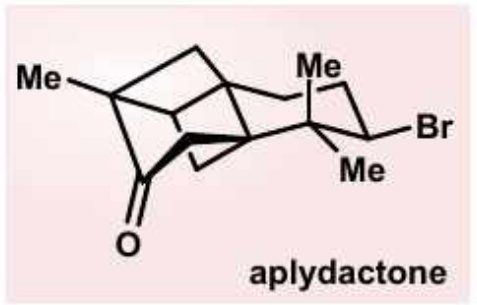


Construction of molecular skeleton

Chemical reviews **2016**, *116*, 8912.

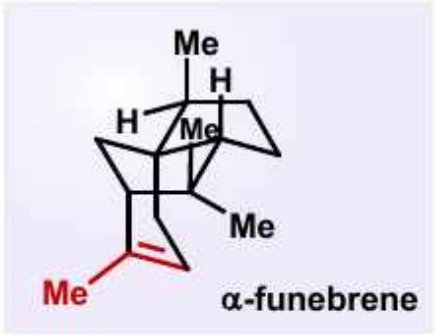
Wonderful Chemoselectivity and Thermodynamic selectivity





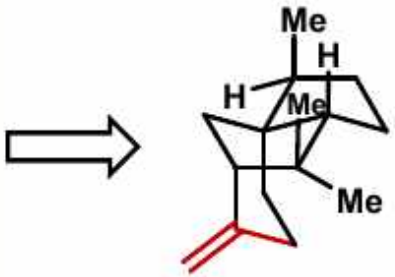
aplydactone

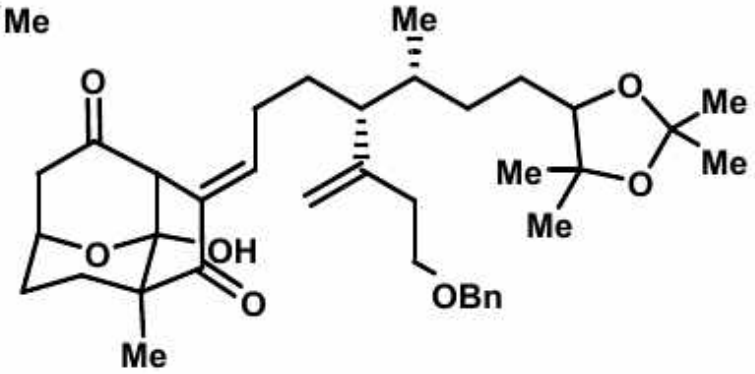
Construction of chiral centers with thermodynamic selectivity

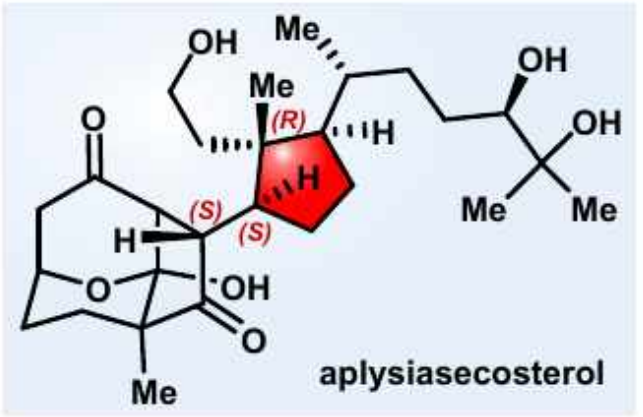


α-funebrene

Mild Condition







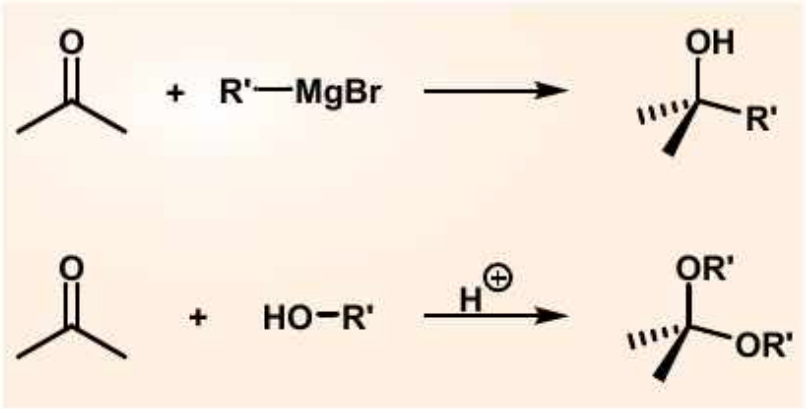
aplysiasecosterol

Green, S. A.; Crossley, S. W.; Matos, J. L.; Vásquez-Céspedes, S.; Shevick, S. L.; Shenvi, R. A.
Accounts of chemical research **2018**, *51*, 2628.

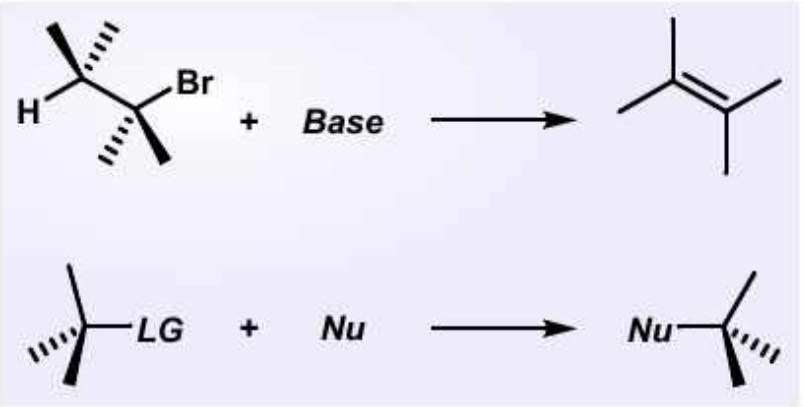
What is the meaning of Metal hydride hydrogen atom transfer reactions

■ Polarity Matching Principle

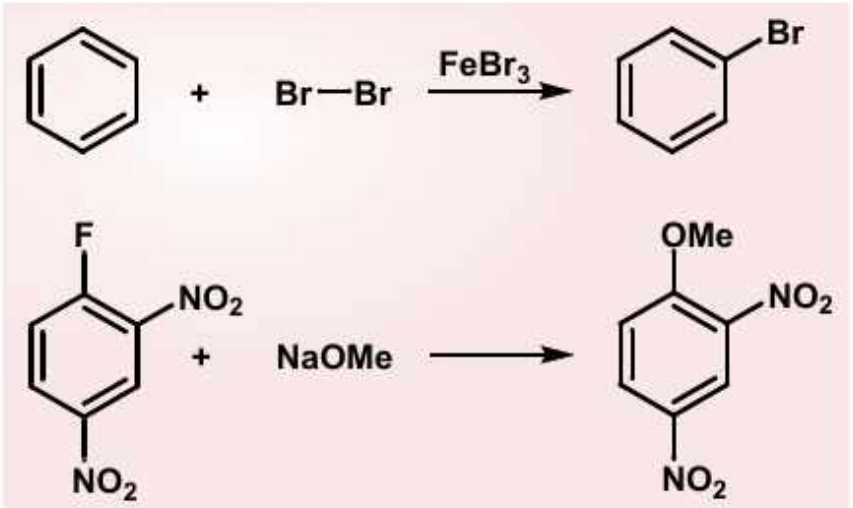
In Carbonyl addition



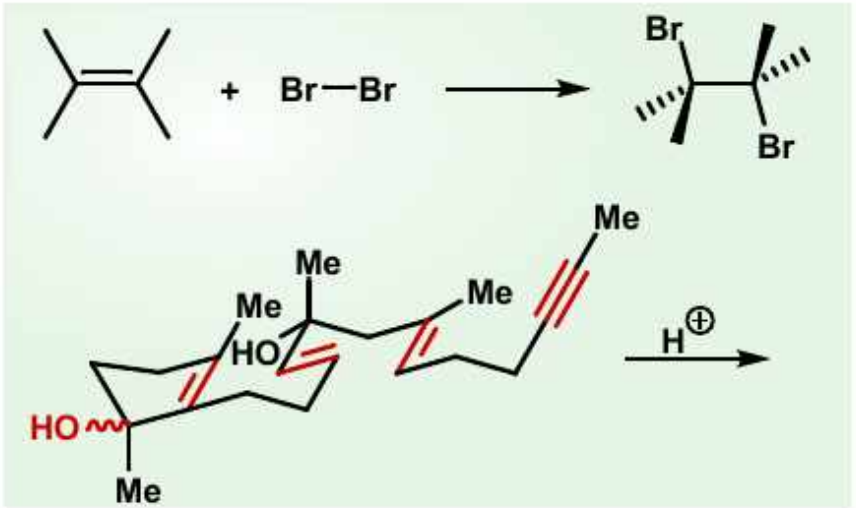
In Nucleophilic substitution at saturated carbon



In Aromatic substitution

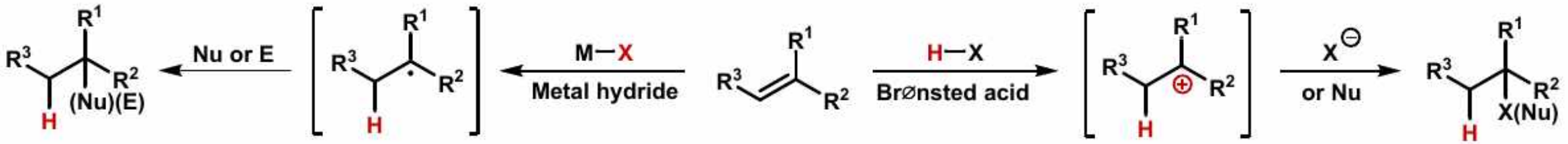


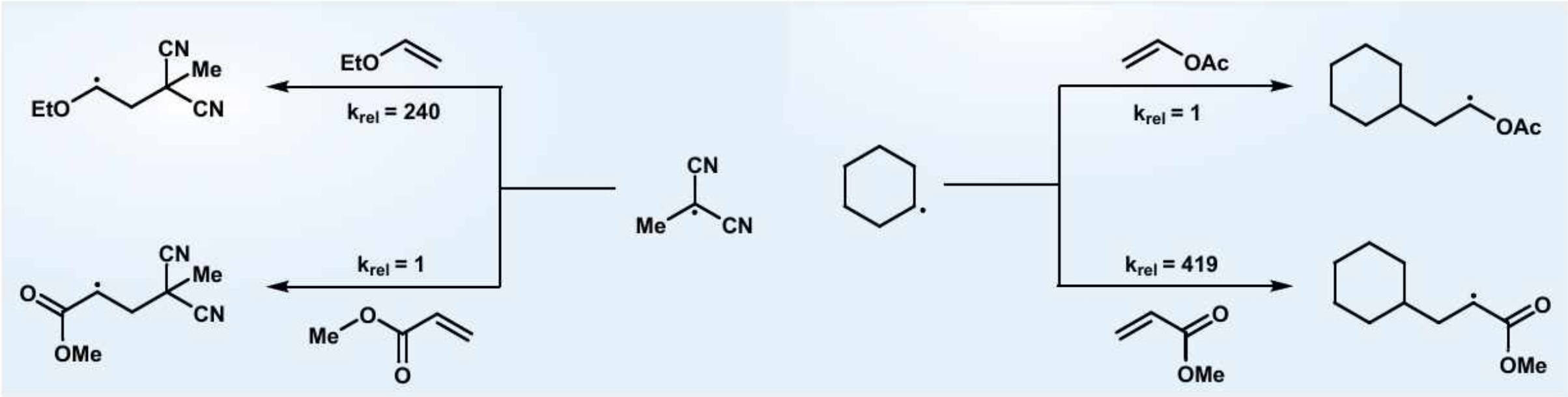
In alkene addtion



What is the meaning of Metal hydride hydrogen atom transfer reactions

■ Polarity Matching Principle in radical reaction

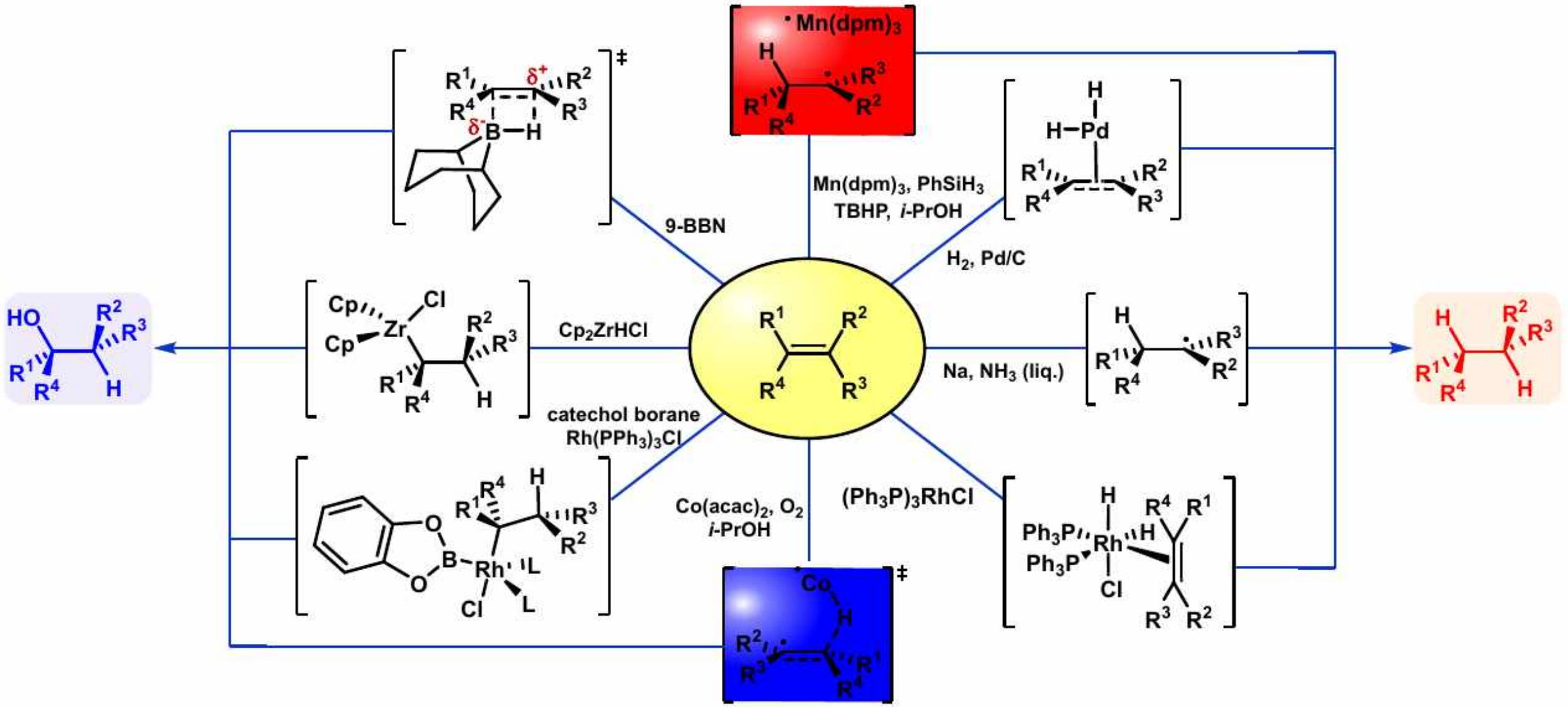




Green, S. A.; Crossley, S. W.; Matos, J. L.; Vásquez-Céspedes, S.; Shevick, S. L.; Shenvi, R. A. *Accounts of chemical research* **2018**, *51*, 2628.

What is the meaning of Metal hydride hydrogen atom transfer reactions

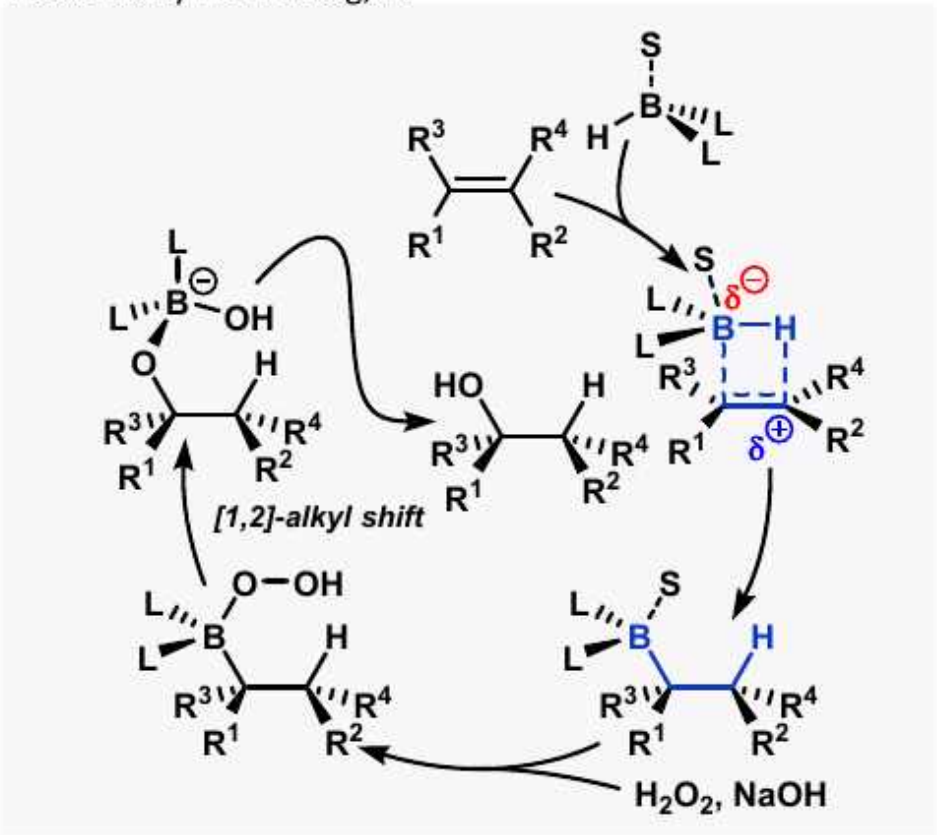
■ The mechanism and intermediate for Alkenes



What is the meaning of Metal hydride hydrogen atom transfer reactions

Hydroboration of alkene with synergistic mechanism

Provided by Prof. Wang, B.



Bonding and Backbonding

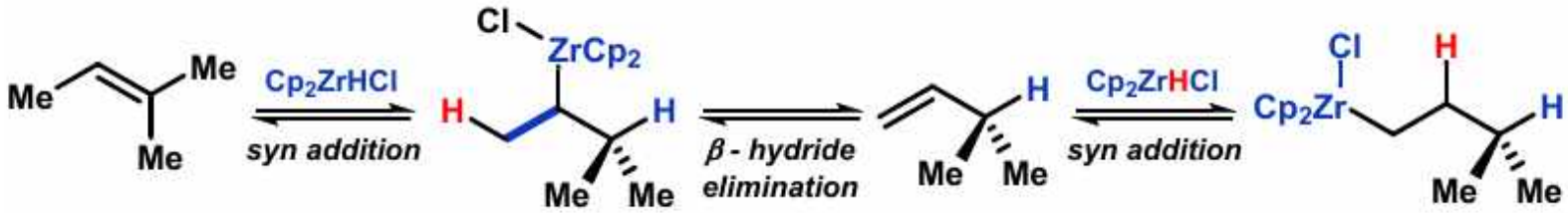
Provided by Prof. Wang, B.



■ The difference between MH-HAT and normal hydriding

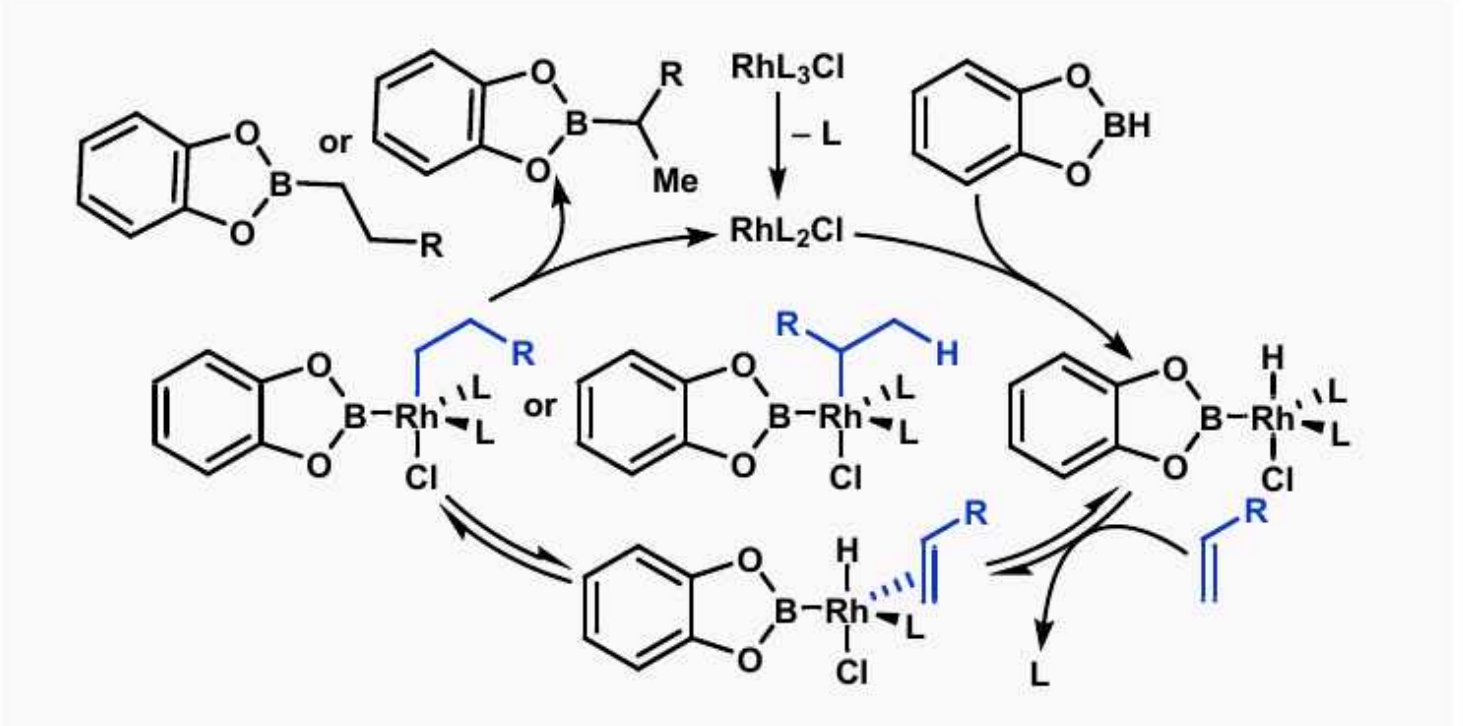
Thermodynamic control & Coordination model

Angewandte Chemie International Edition in English **1976**, *15*, 333.



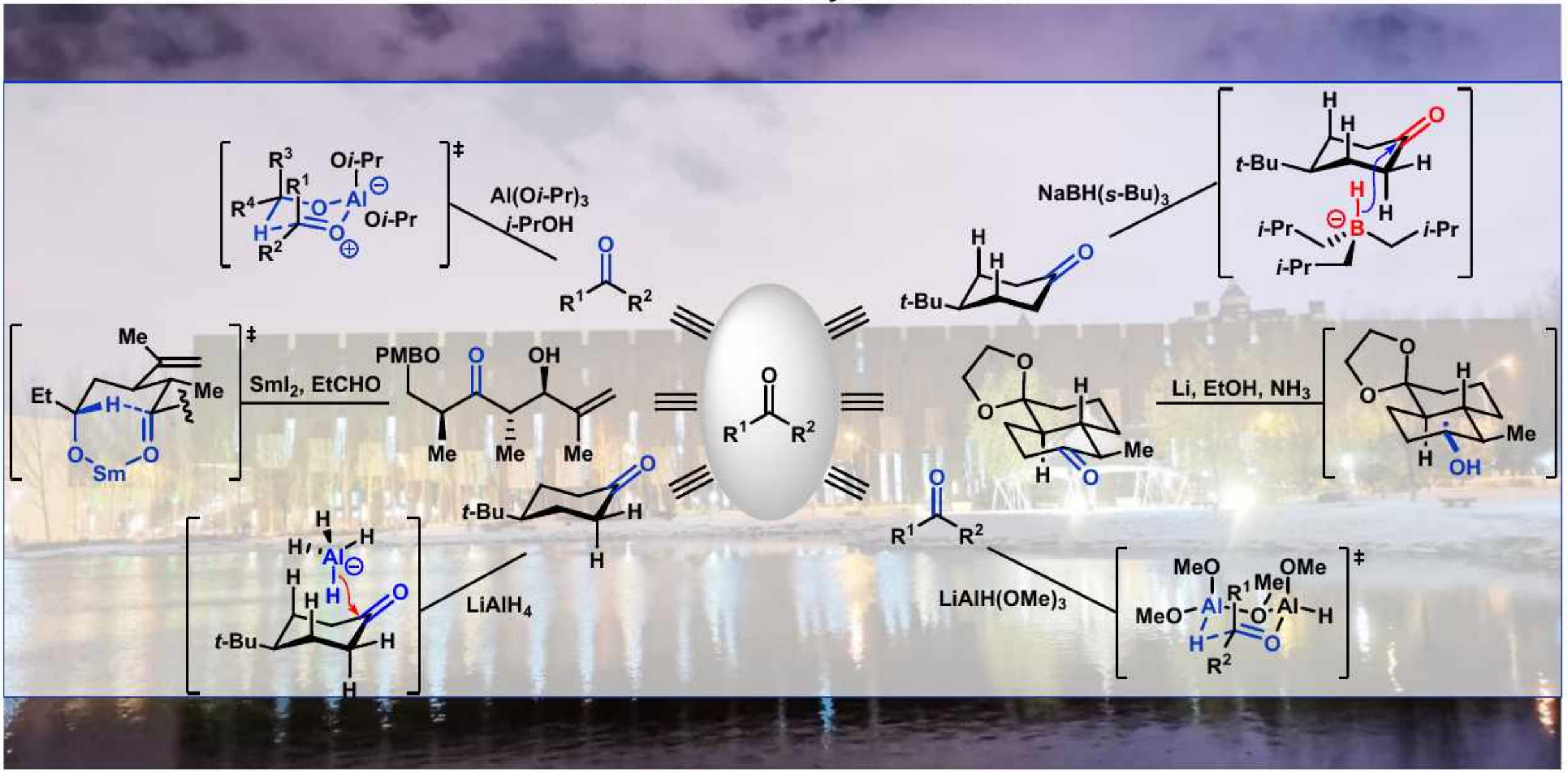
Via the coordination of the double bond and the matal

Journal of the American Chemical Society **1988**, *110*, 6917.



What is the meaning of Metal hydride hydrogen atom transfer reactions

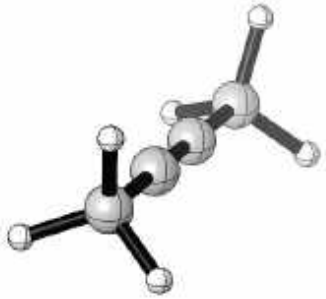
- How about the aldehyde and ketone



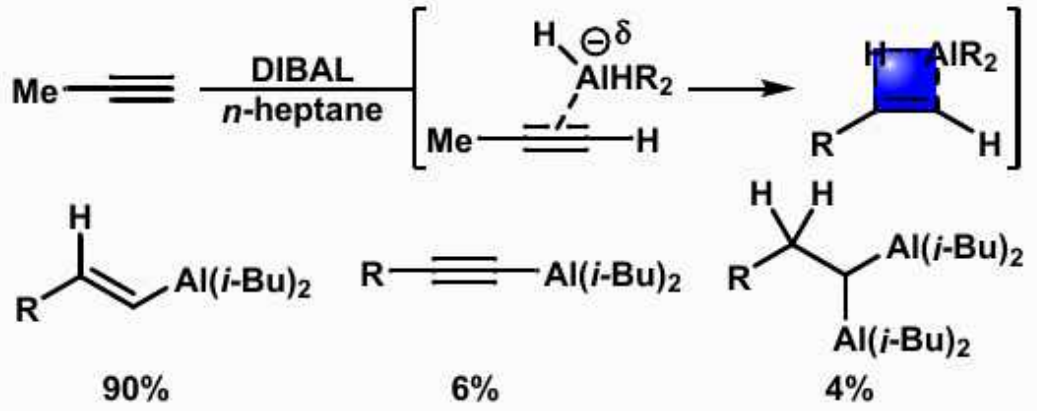
What is the meaning of Metal hydride hydrogen atom transfer reactions

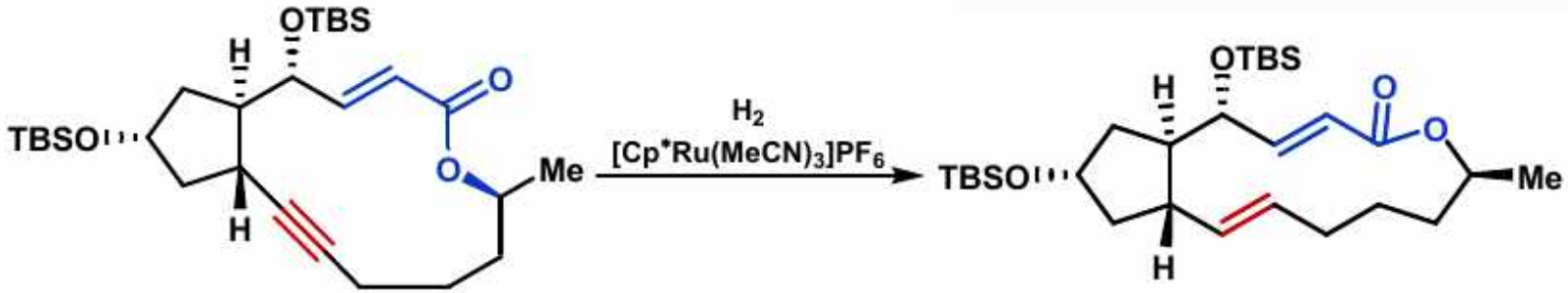
- Why it can not be used in Hydrogenation of alkyne



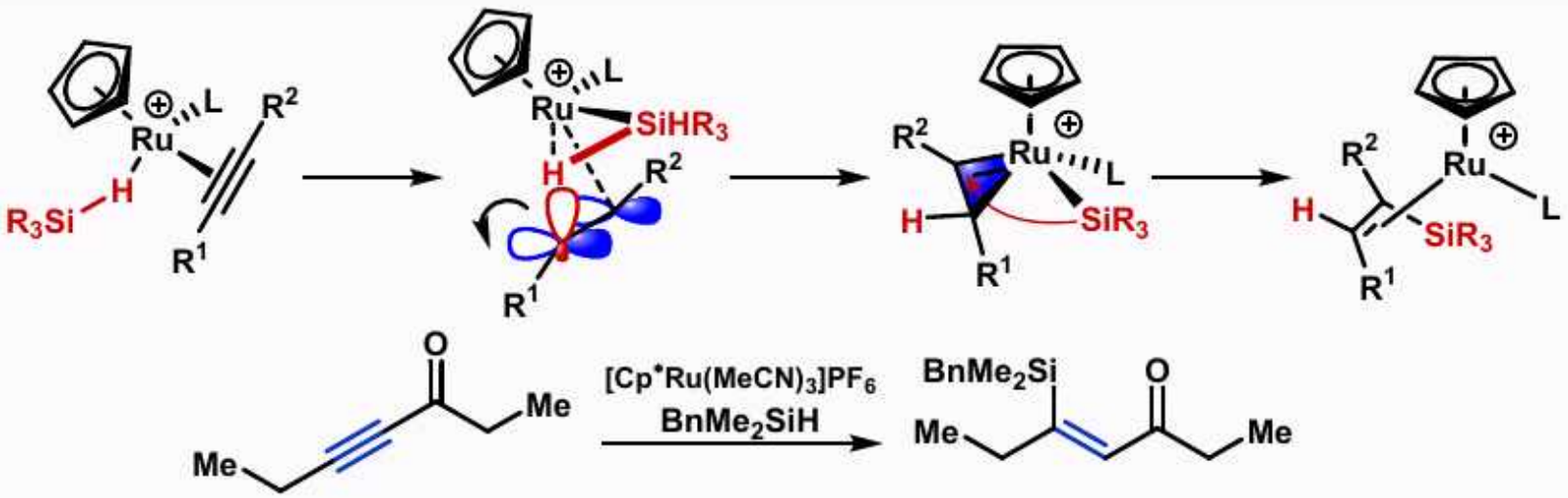


***trans*-Hydrogenation of alkyne via aluminum aluminum compounds**
Provided by Prof. Wang, B





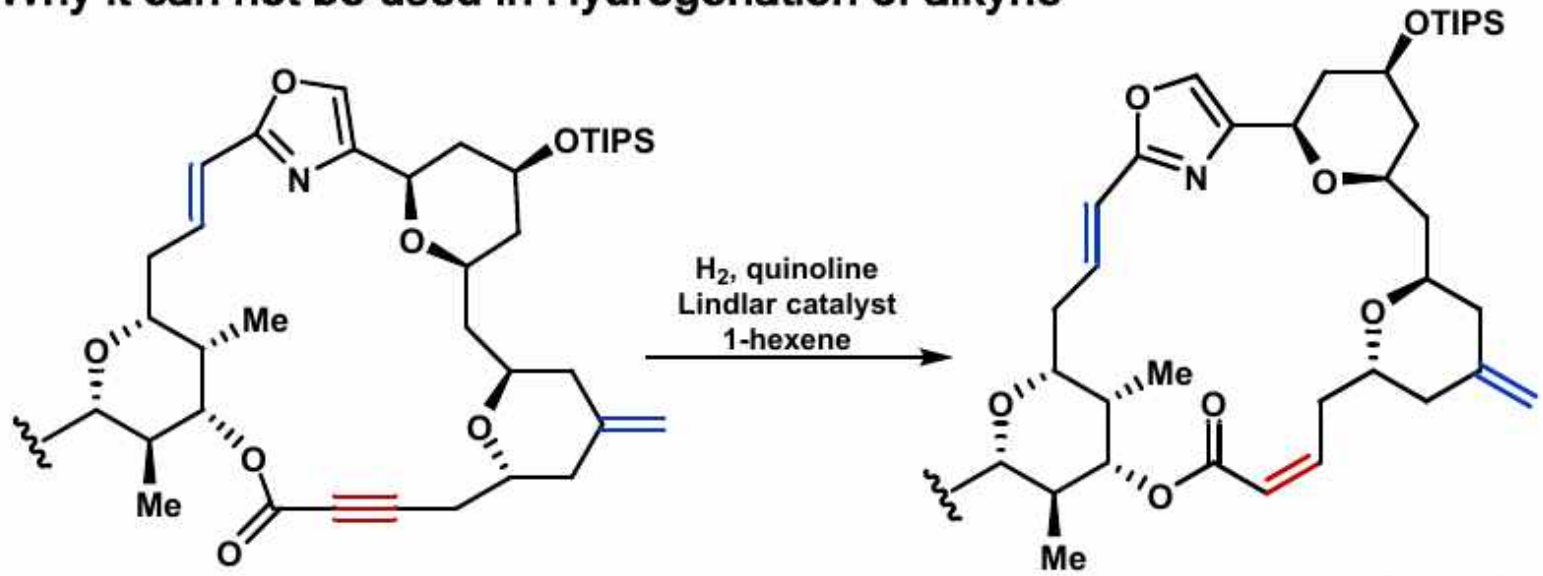
***trans*-Hydrogenation of alkyne**
Angewandte Chemie International Edition **2015**, *54*, 3978.



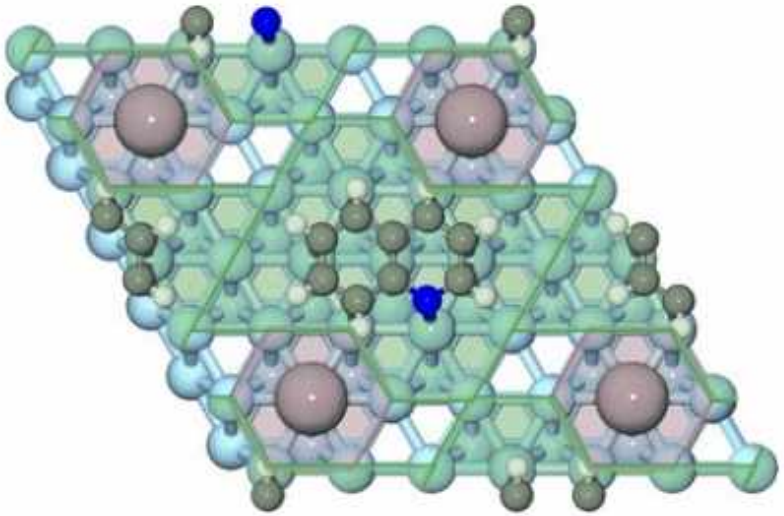
Mechanism involving a direct three-component addition
Journal of the American Chemical Society **2005**, *127*, 17644.

What is the meaning of Metal hydride hydrogen atom transfer reactions

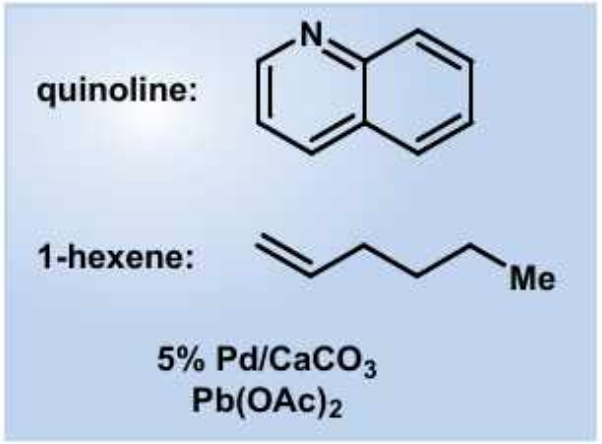
- Why it can not be used in Hydrogenation of alkyne

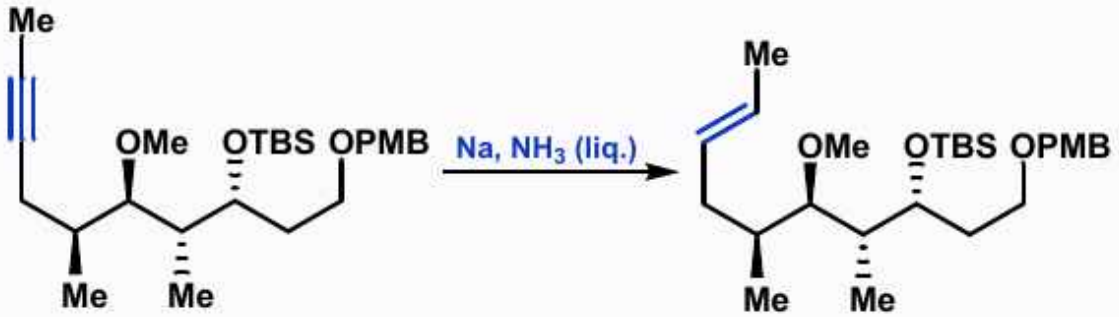


***cis*-Semihydrogenation using Lindlar's catalyst**
Journal of the American Chemical Society **2000**, *122*, 10033.

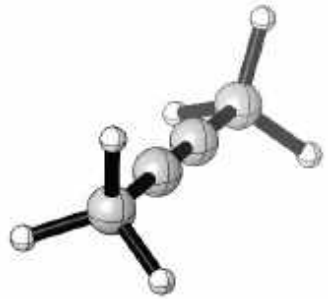


Schematic representation of the mosaic generated by the superposition of the Pb exclusion areas (*pink*) and the quinoline ones (*gray*).
Angewandte Chemie International Edition **2015**, *54*, 3978.





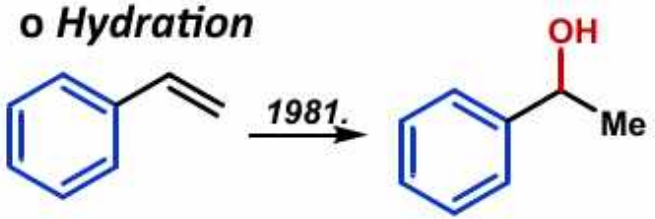
***trans*-Semihydrogenation using Sodium-liquid reduction**
Organic Letters **2003**, *5*, 265.



Retrosynthetic possibility of Metal hydride hydrogen atom transfer reactions

■ Formation of Carbon—Oxygen Bonds

o *Hydration*



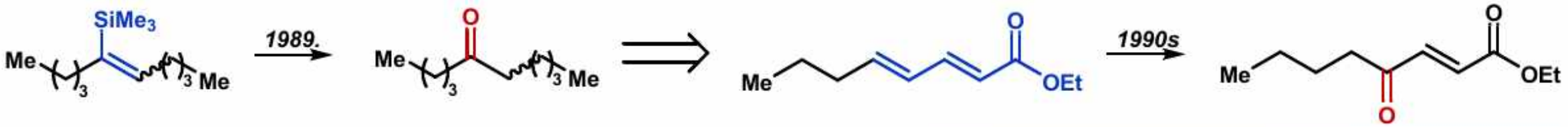
o *Hydroperoxidation*

——*Story begin with oxygen*

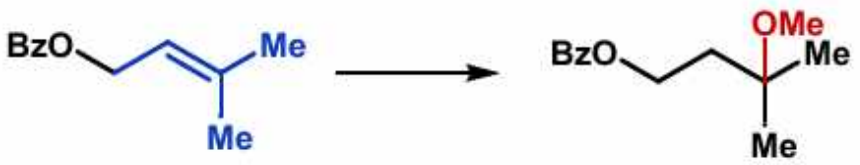
Retrosynthetic possibility of Metal hydride hydrogen atom transfer reactions

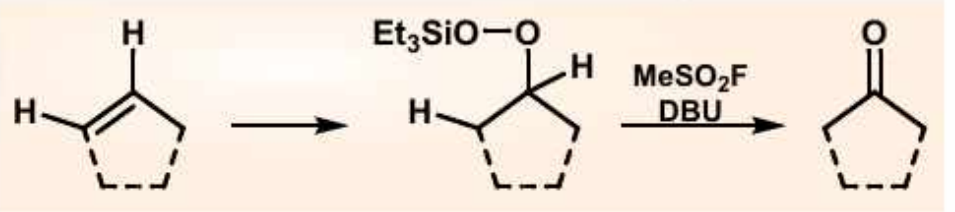
■ Formation of Carbon—Oxygen Bonds

o *Ketone Formation*

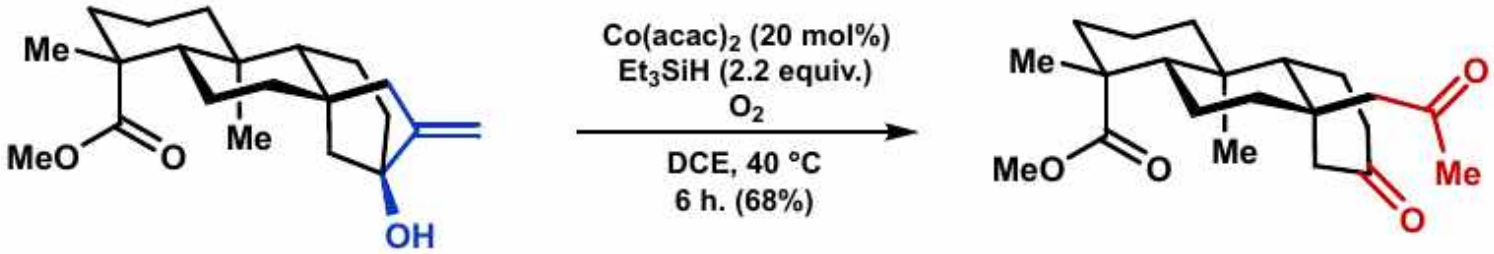


o *Hydroalkoxylation*

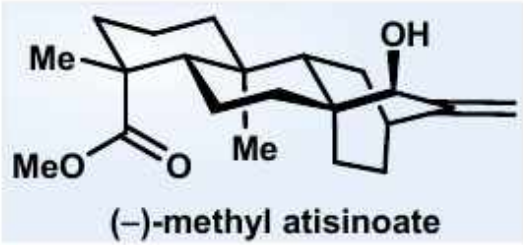




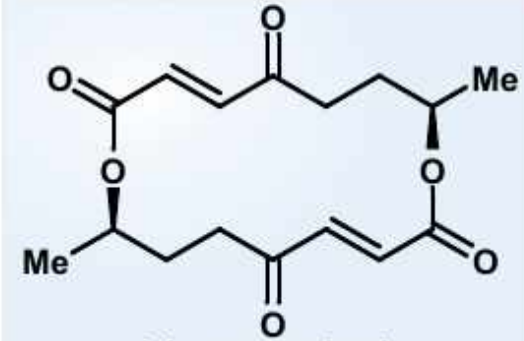
o *C-O Bond Formation in Natural Products*







(–)-methyl atisinoate

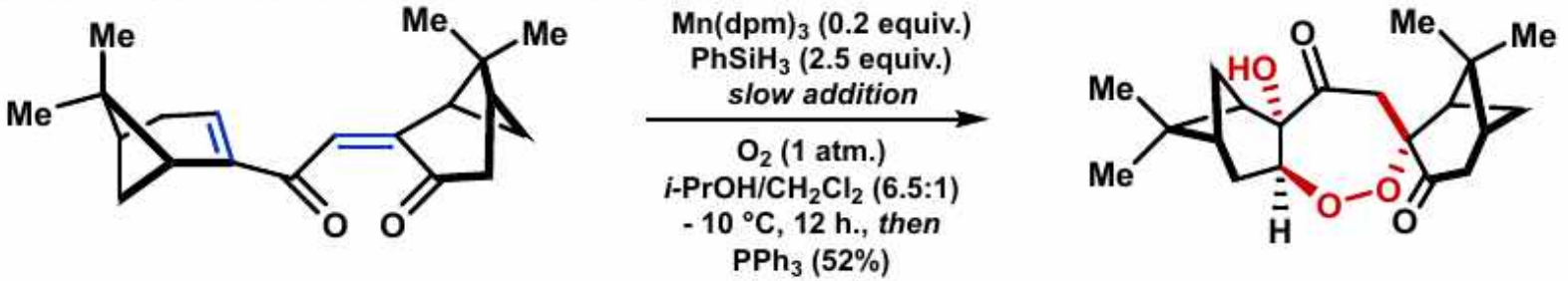


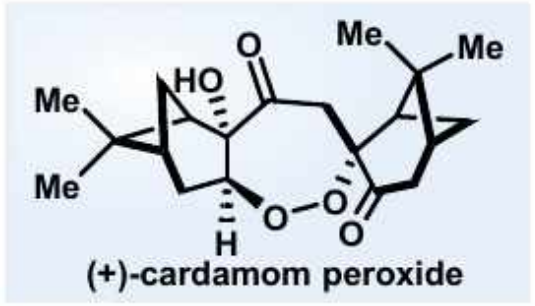
(–)-pyrenophorin

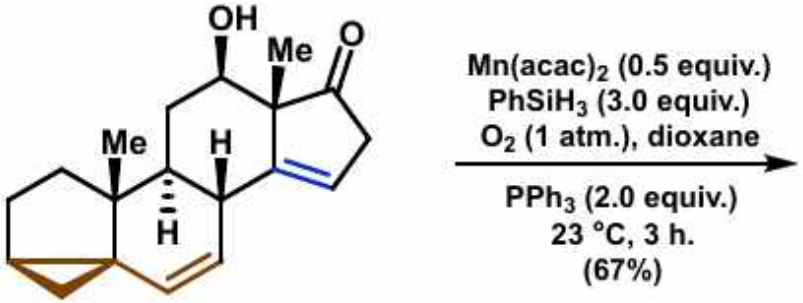
Retrosynthetic possibility of Metal hydride hydrogen atom transfer reactions

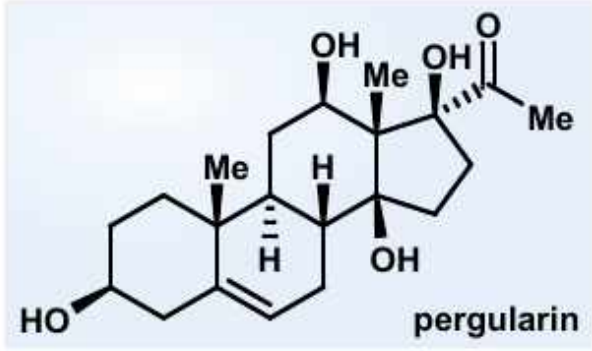
■ Formation of Carbon—Oxygen Bonds

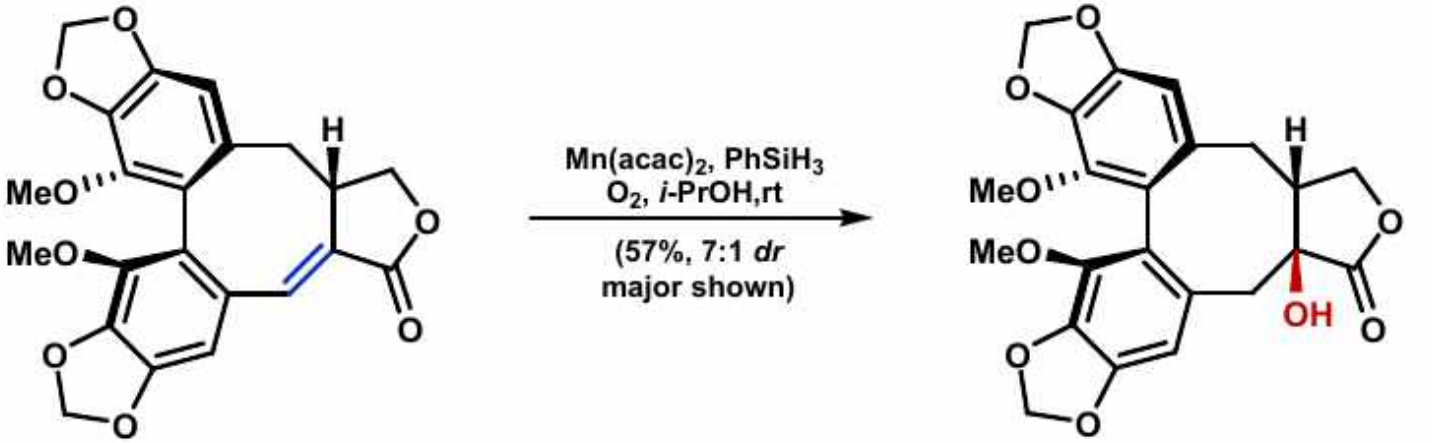
o *C-O Bond Formation in Natural Products*

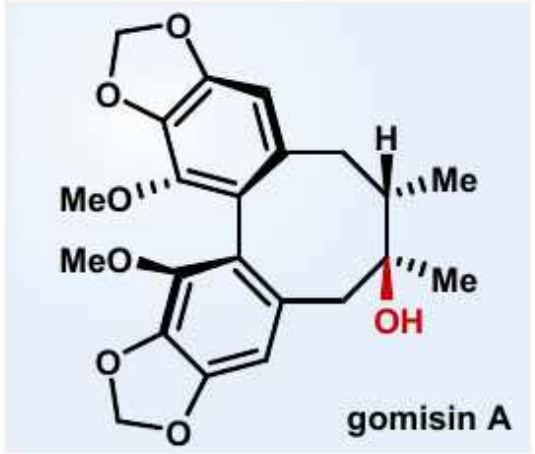










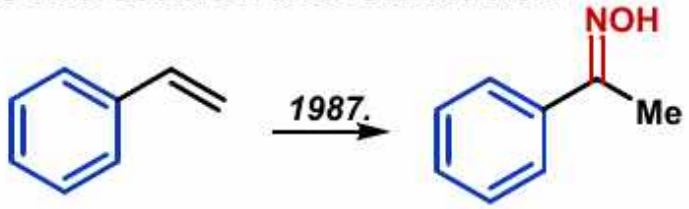


Retrosynthetic possibility of Metal hydride hydrogen atom transfer reactions

■ Formation of Carbon—Nitrogen Bonds

o Nitrosation and Oximation

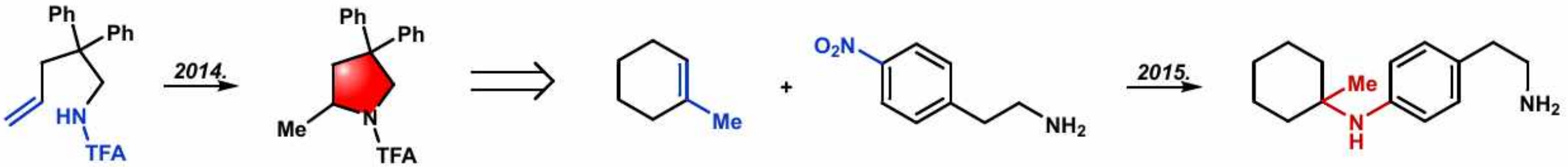
——What about nitric oxide



Retrosynthetic possibility of Metal hydride hydrogen atom transfer reactions

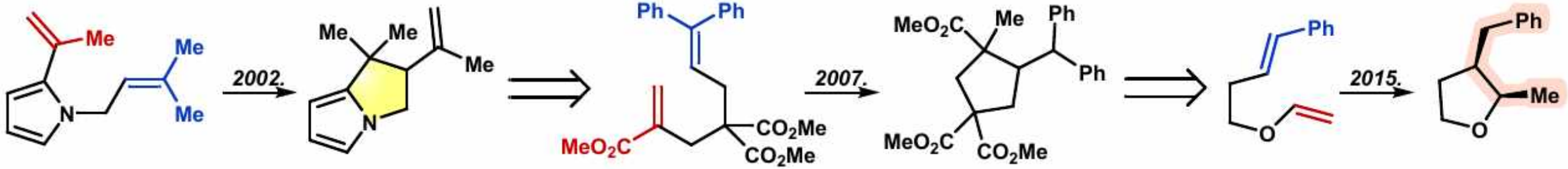
■ Formation of Carbon—Nitrogen Bonds

o *Hydroamination*

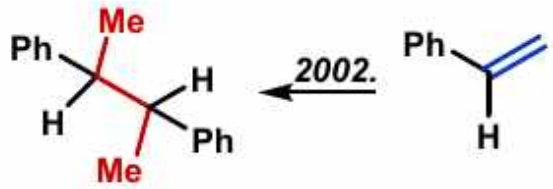


■ Formation of Carbon—Carbon Bonds

o *Reductive Carbocyclization*



o *Intermolecular Reductive Dimerization*



o *Hydrocyanation*



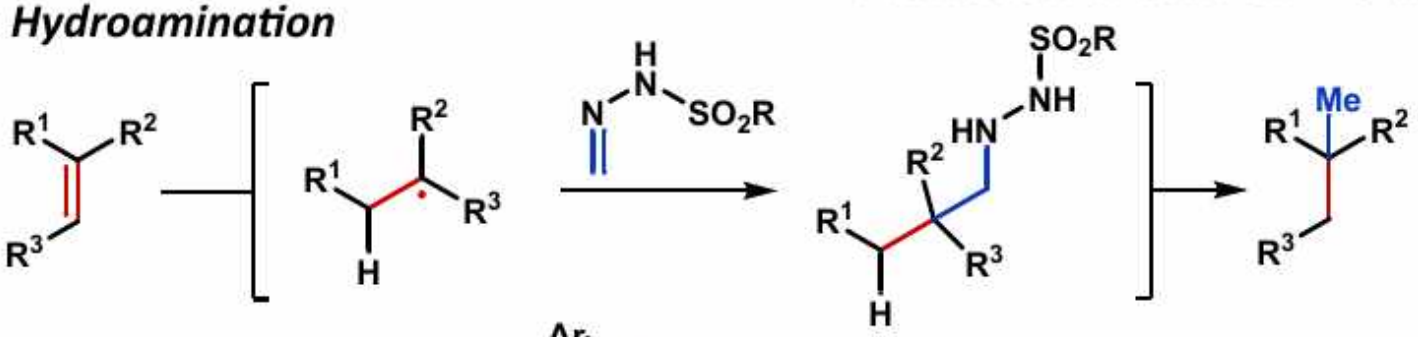
o *Conjugate Addition*

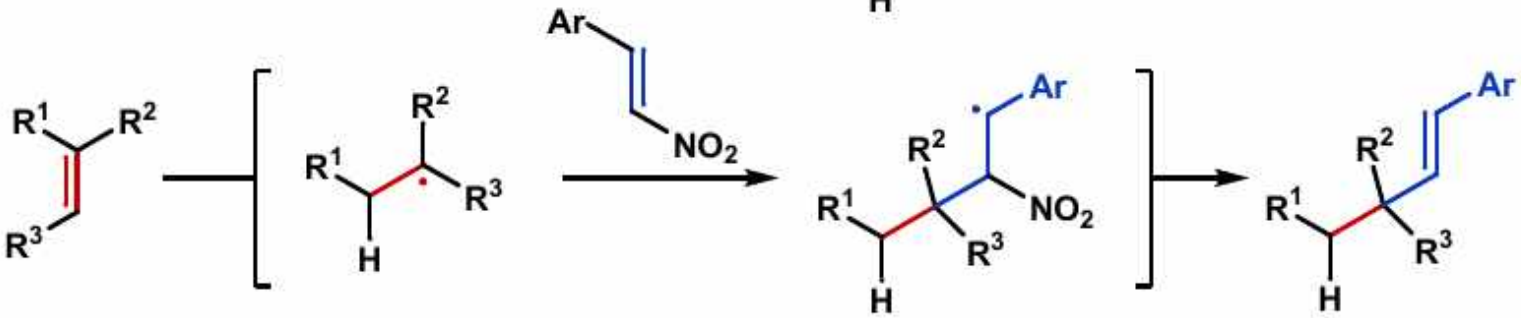


Retrosynthetic possibility of Metal hydride hydrogen atom transfer reactions

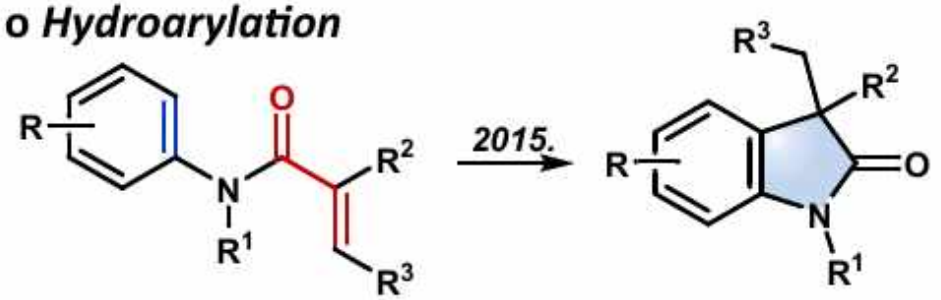
■ Formation of Carbon—Carbon bonds

o Hydroamination

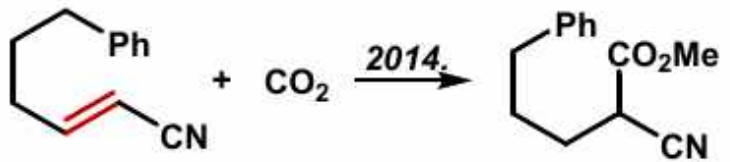




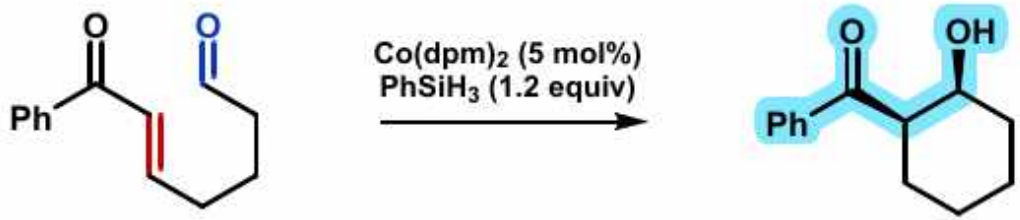
o Hydroarylation



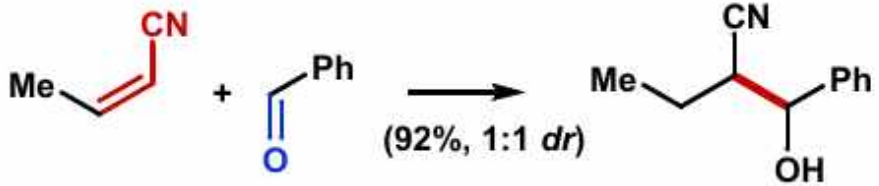
o Hydrocarboxylation

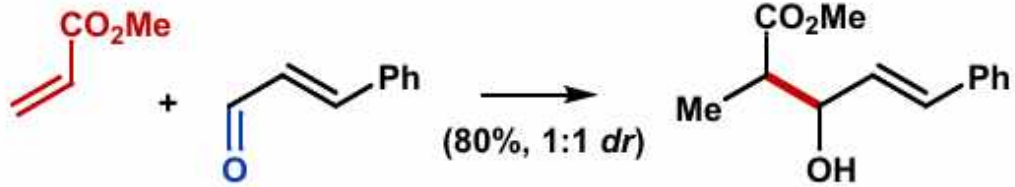


o Reductive Coupling of α,β-Unsaturated Compouds with Carbonyls





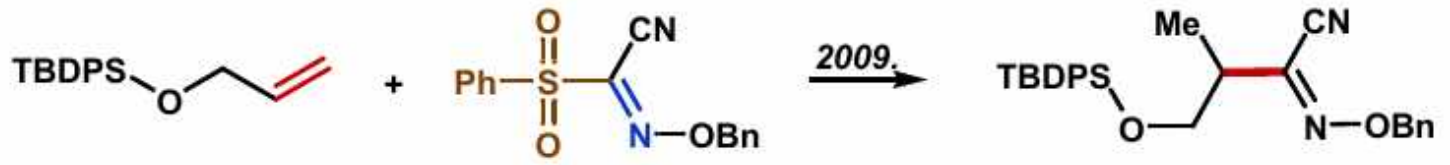


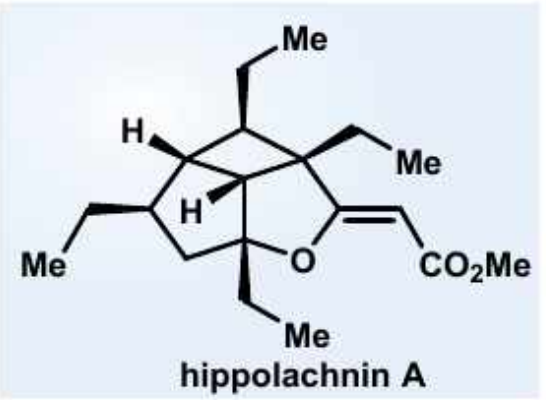


Retrosynthetic possibility of Metal hydride hydrogen atom transfer reactions

■ Formation of Carbon—Carbon bonds

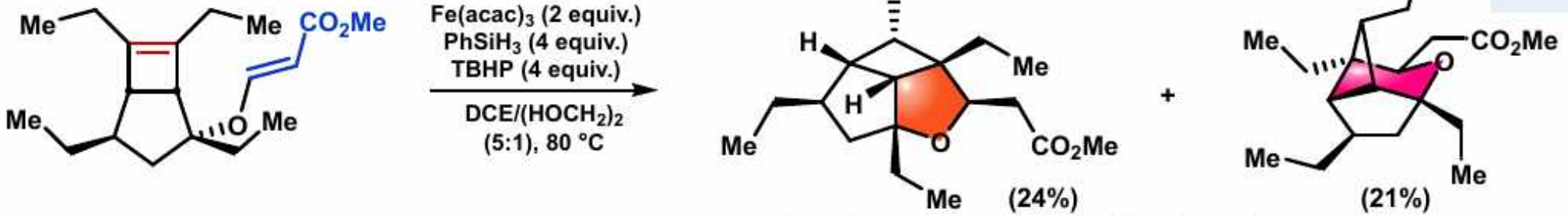
o *Reductive Addition of Alkenes to Sulfonyl Oximes*





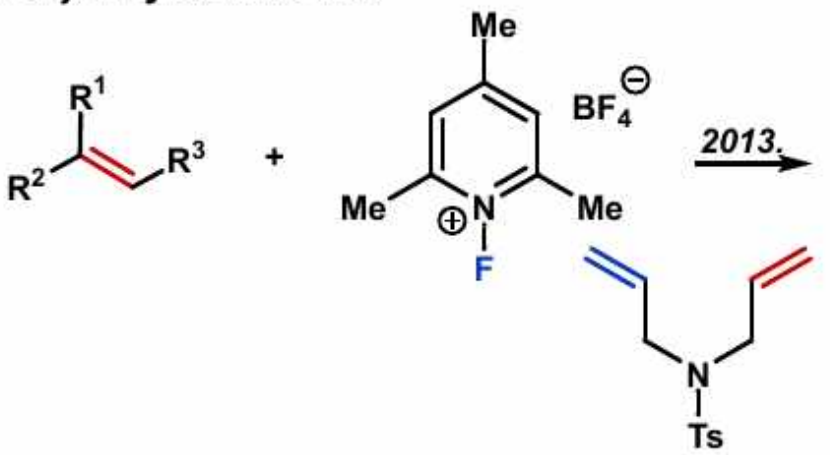
hippolachnin A

o *C-C Bond Formation in Natural Products*

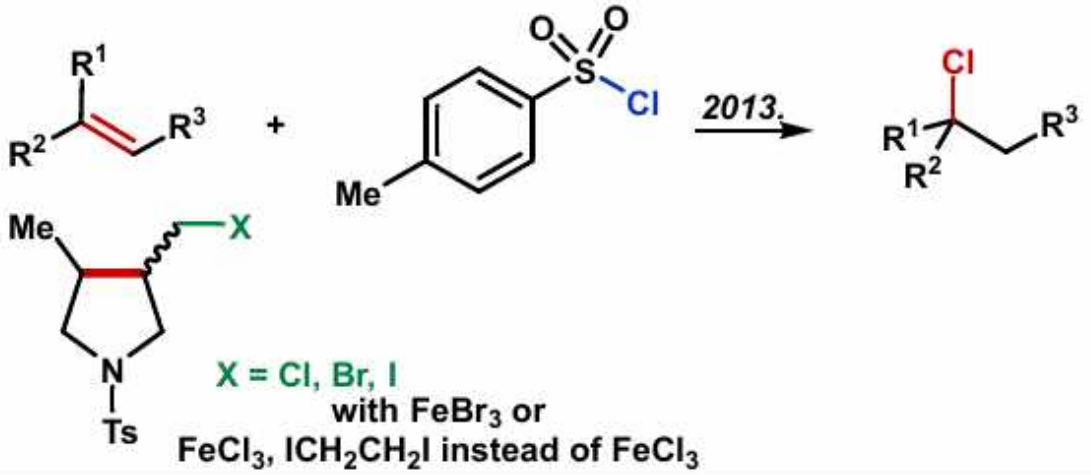


■ Formation of Carbon—X Bonds (X = Halogen, S, SE)

o *Hydrofluorination*



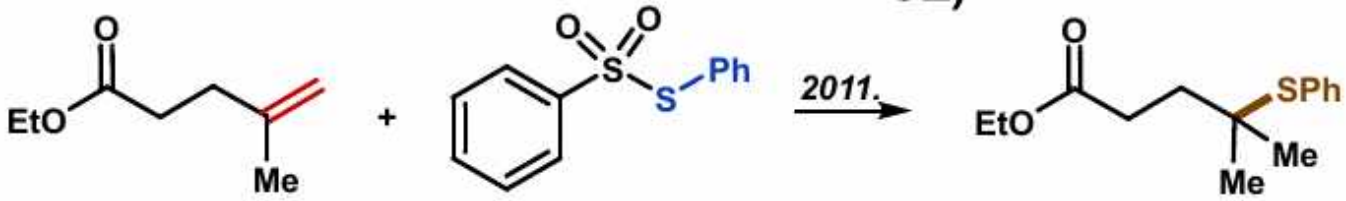
o *Hydrochlorination/Bromination/Iodination*



Retrosynthetic possibility of Metal hydride hydrogen atom transfer reactions

■ Formation of Carbon—X Bonds (X = Halogen, S, SE)

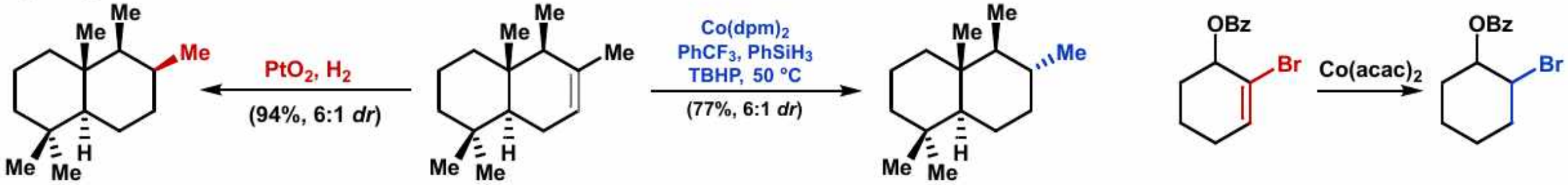
o *Hydrochalcogenation*



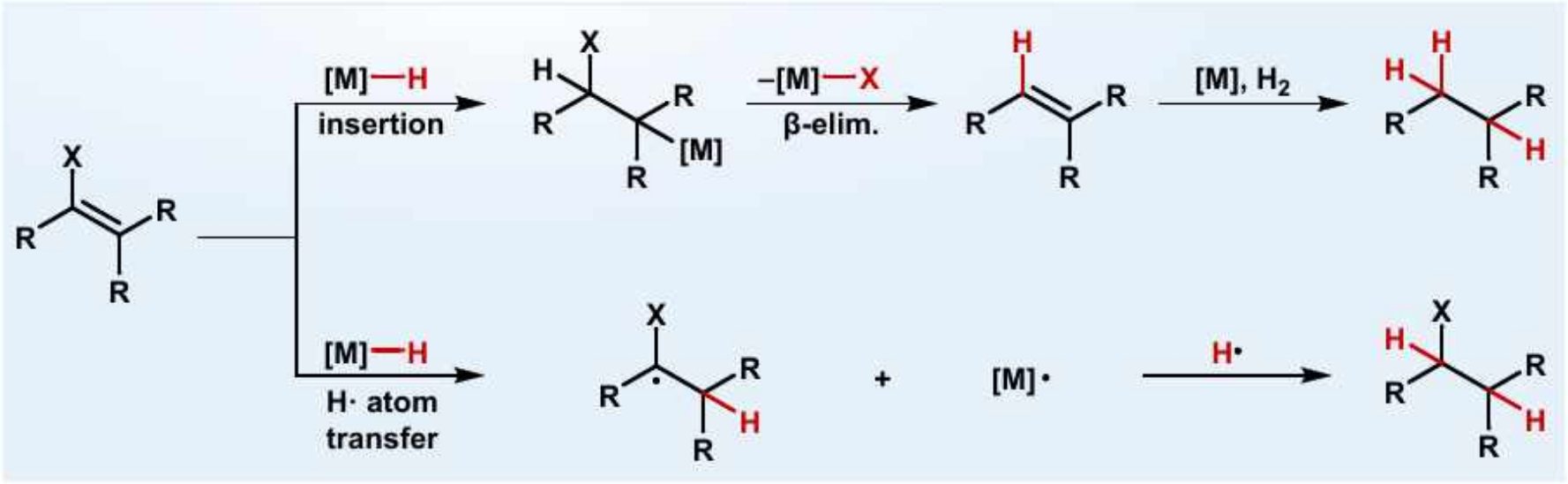
Can also use —SePh instead of —SPh to get Se Bonds

■ Hydrogenation and Isomerization

o *Hydrogenation*



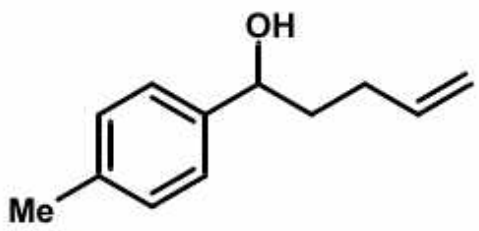
Great Chemoselectivity



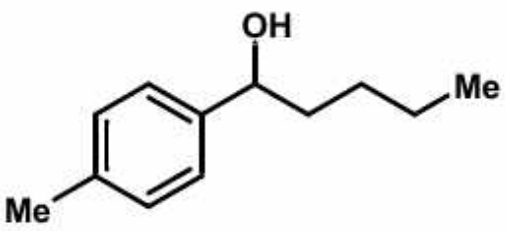
Retrosynthetic possibility of Metal hydride hydrogen atom transfer reactions

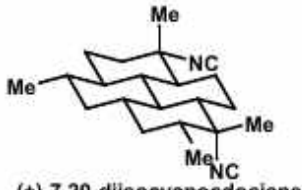
■ Hydrogenation and Isomerization

o *Reaction proceeds through a polar mechanism, not through radicals.*



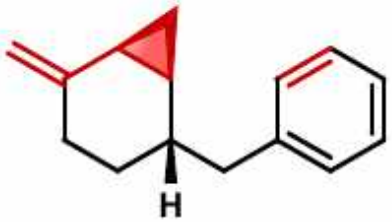
$Fe(OTf)_3$ (10 mol%)
$NaBH_4$
EtOH, rt, 6 h.





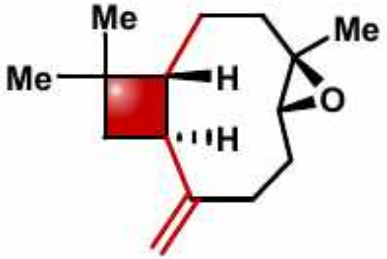
(+)-7,20-diisocyanoadociane

o *Isomerization*

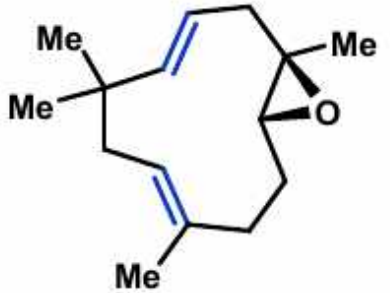


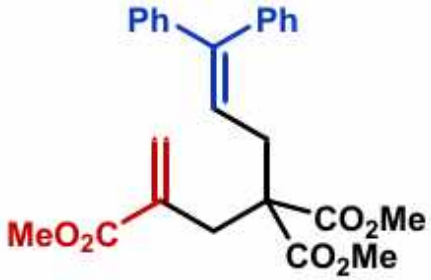
$Co(Salen^{tBu,tBu})Cl$
(1-10 mol%)
$PhSiH_3$ (1-50 mol%)
PhH, Argon atm



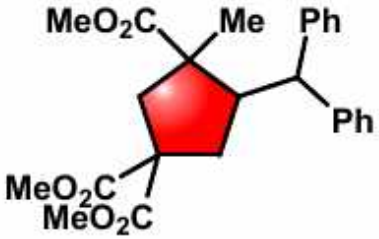


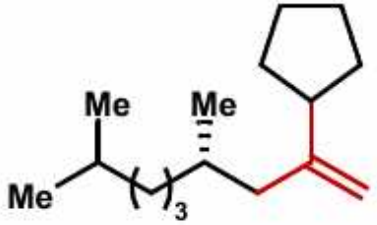
$Co(Salen^{tBu,tBu})Cl$
(1-10 mol%)
$PhSiH_3$ (1-50 mol%)
PhH, Argon atm



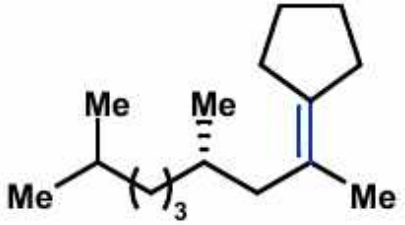


$CpCr(CO)_3H$ (7 mol%)
H_2 (30 psi)
C_6D_6 [0.30 M]
50 °C, 1.5 days

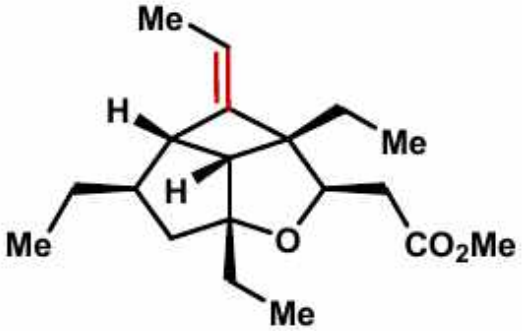




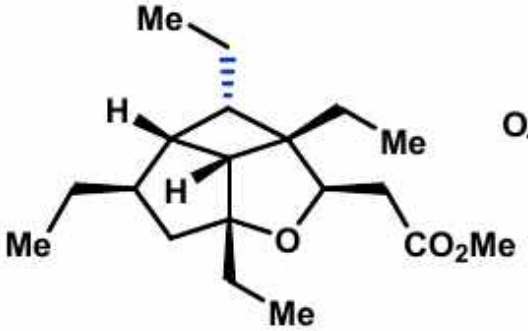
$Co(Salen^{tBu,tBu})Cl$
(1-10 mol%)
$PhSiH_3$ (1-50 mol%)
PhH, Argon atm

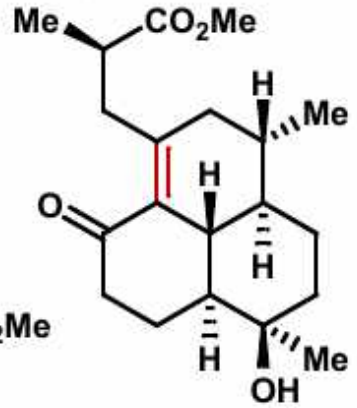


o *In Natural Products*

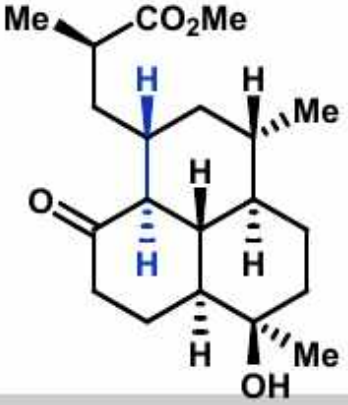


$Mn(dpm)_3$ (30 mol%)
$PhSiH_3$ (3 equiv.)
TBHP (4 equiv.)
i-PrOH (0.05 M)
(75%; > 99:1 *dr*)





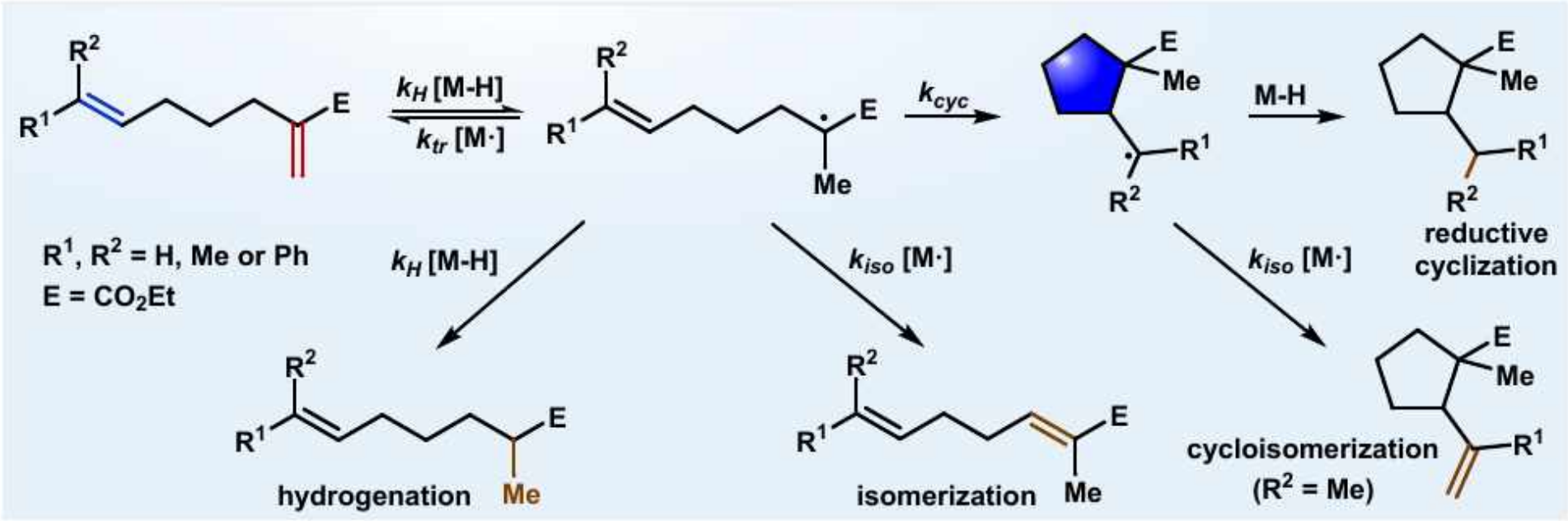
$Mn(dpm)_3$, TBHP
$PhSiH_3$, *i*-PrOH
(3:1 *dr*
51% one isomer)



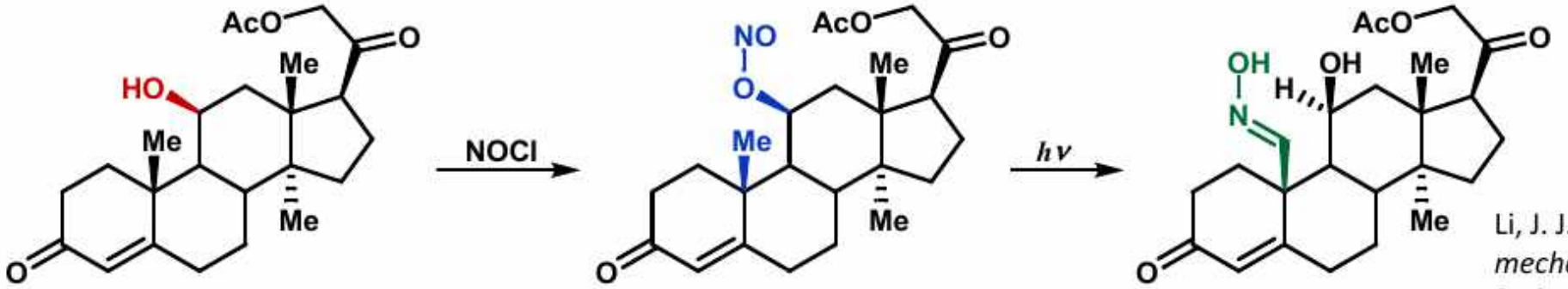
Retrosynthetic possibility of Metal hydride hydrogen atom transfer reactions

■ Hydrogenation and Isomerization

o *Isomerization*



■ Back to an old and classical reaction

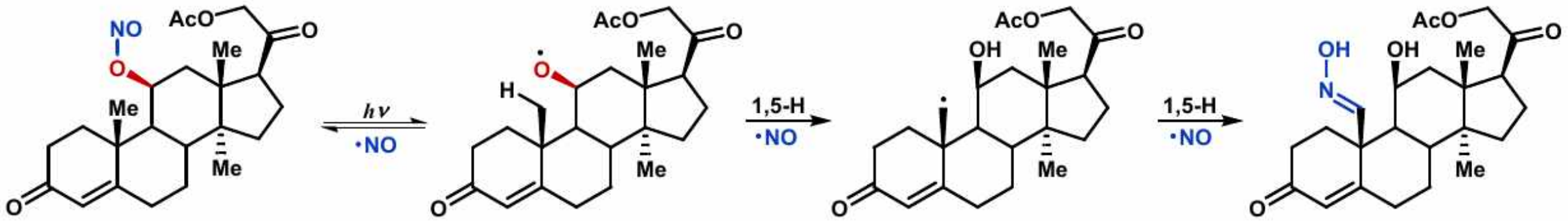


Li, J. J. *Name reactions: a collection of detailed mechanisms and synthetic applications*; Springer Science & Business Media, 2010.

Mechanistic Overview and a more Realistic Situation

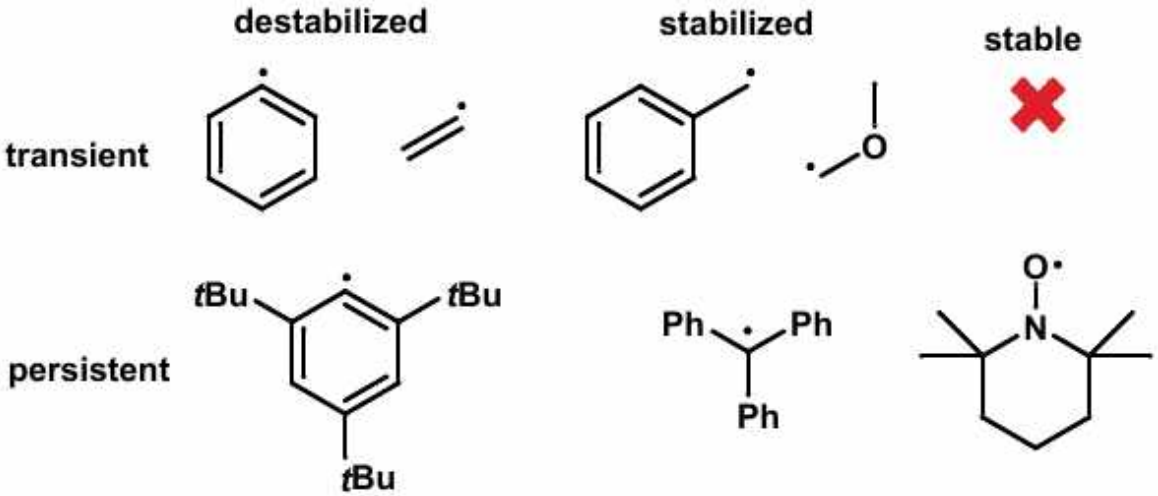
■ The Persistent Radical Effect in Organic Synthesis

Why Barton's reaction can form a oxime

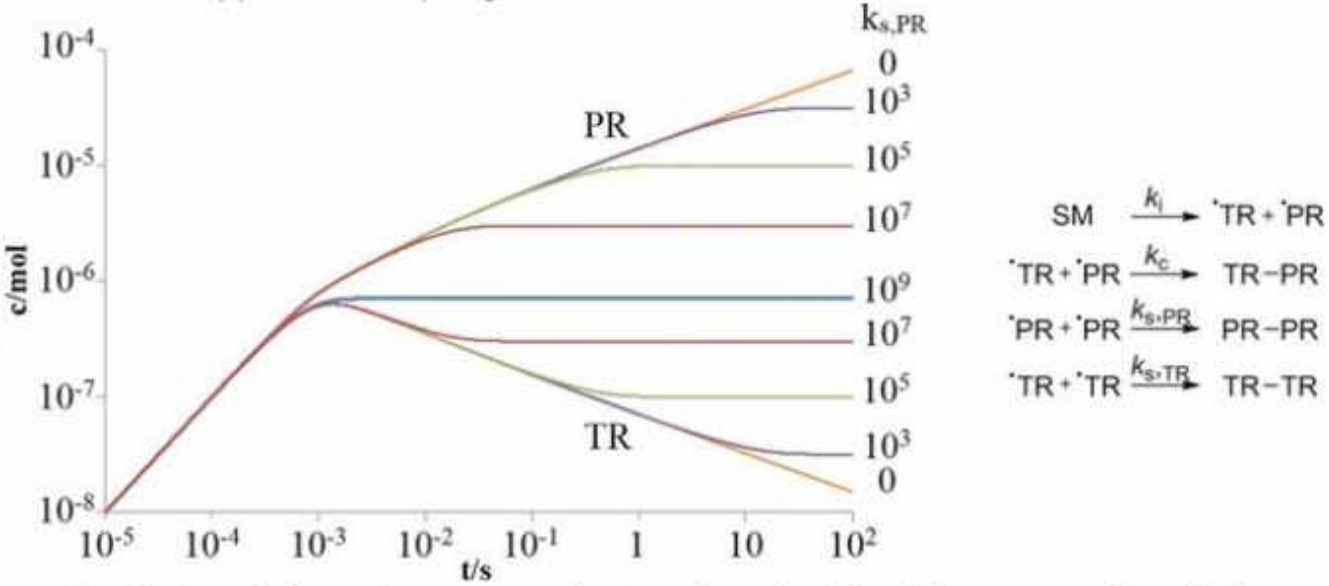


Why Barton's reaction can form a oxime not a C-C bond or N-N bond.
Name reactions: a collection of detailed mechanisms and synthetic applications; Springer Science & Business Media, 2010.

Persistent Radicl Effect



Selected radicals and thermodynamic and kinetic Categorization based on the definitions of Griller and Ingold.
Angewandte Chemie International Edition **2020**, *59*, 74.

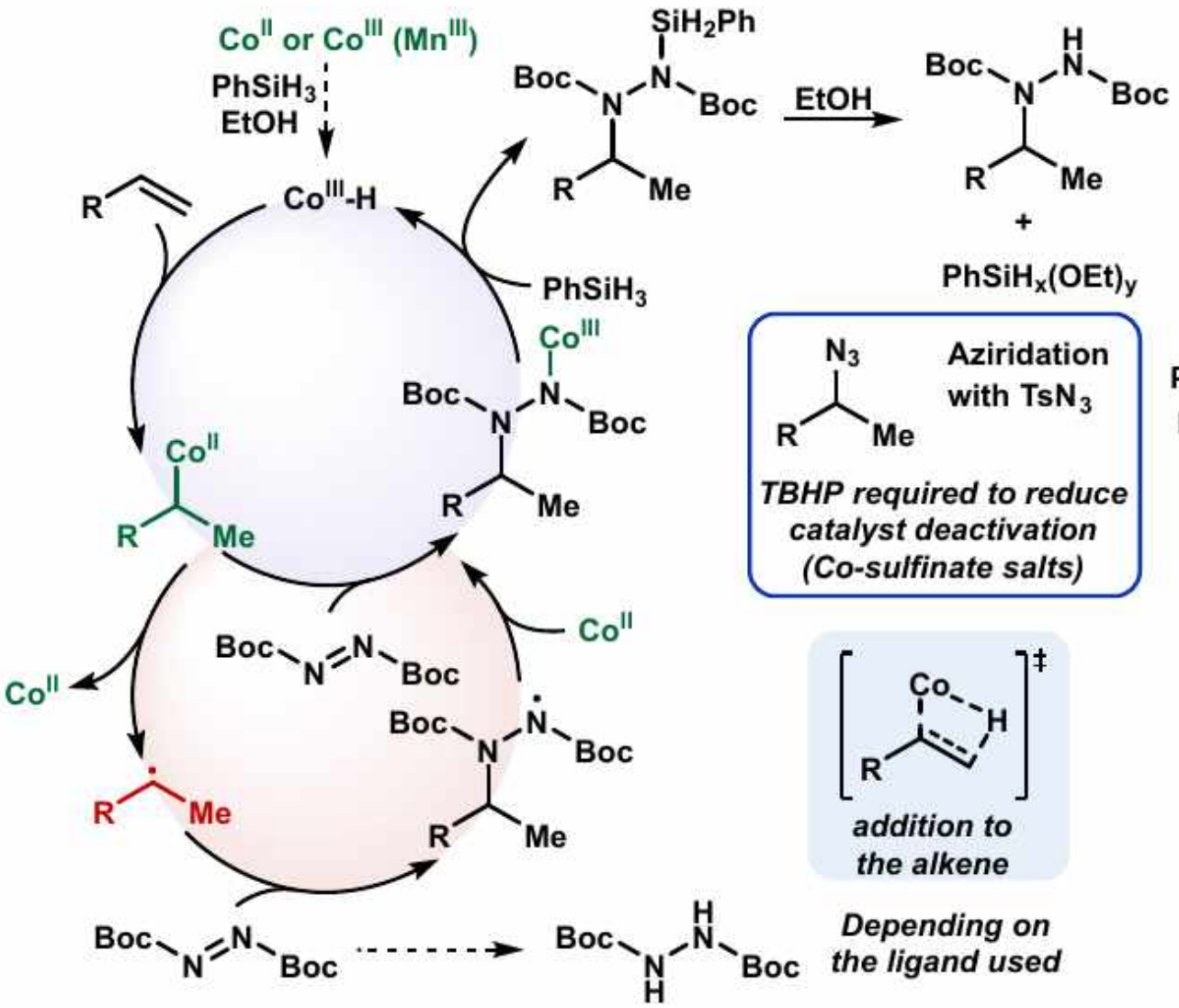


Simulation of dynamic systems obeying the principle of the PRE. Both radicals PR and TR are continuously generated at equal rates
Angewandte Chemie International Edition **2020**, *59*, 74.

Mechanistic Overview and a more Realistic Situation

■ Some mechanistic proposed in the past

Proposed catalytic cycle for the hydrohydrazination and hydroazidation of alkenes



PhSiH$_x$(OEt)$_y$

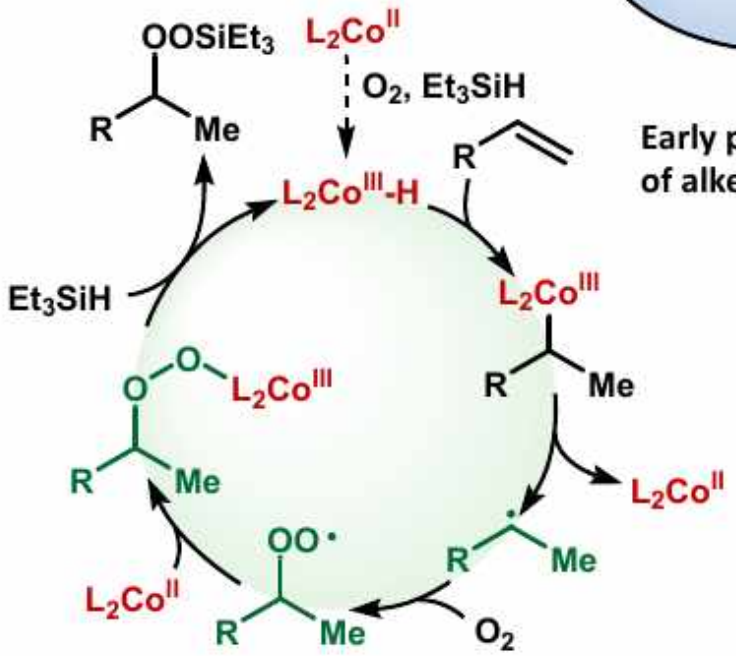
Aziridation with TsN_3

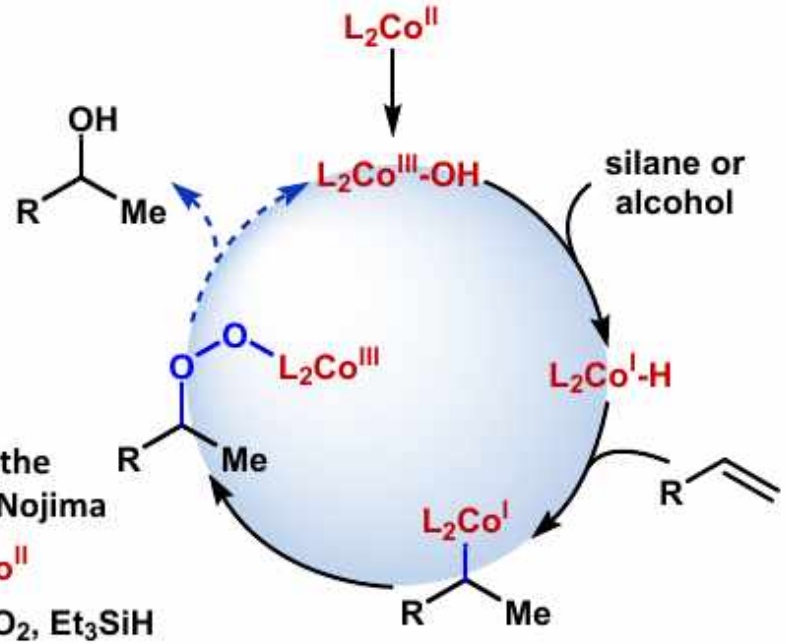
TBHP required to reduce catalyst deactivation (Co-sulfinate salts)

addition to the alkene

Depending on the ligand used

Proposed catalytic cycle for the peroxidation of alkenes by Nojima

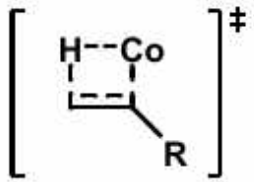




Early proposal for the hydration of alkenes by Mukaiyama.

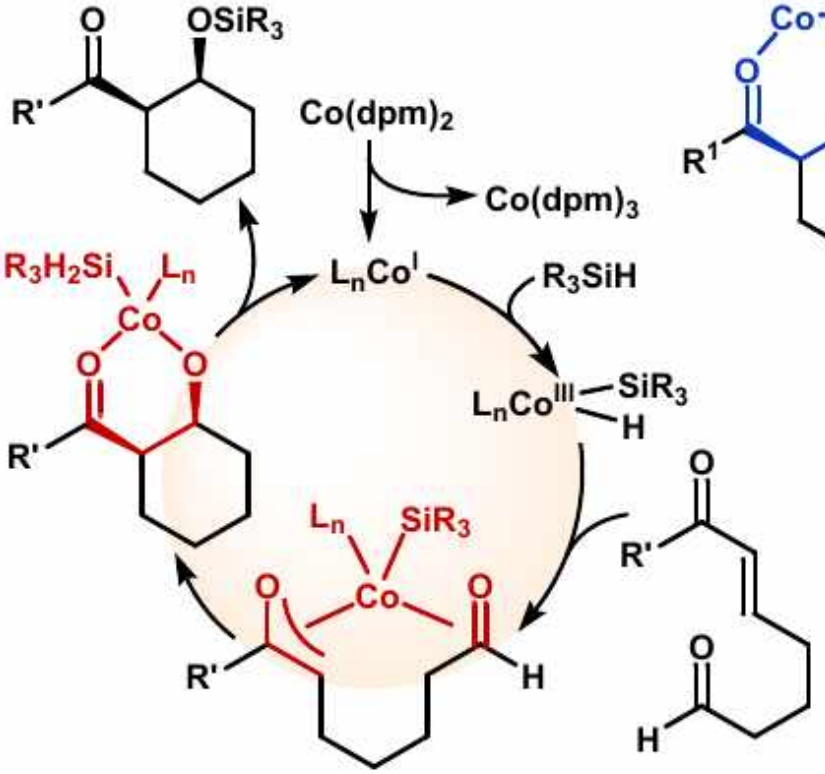
Mechanistic Overview and a more Realistic Situation

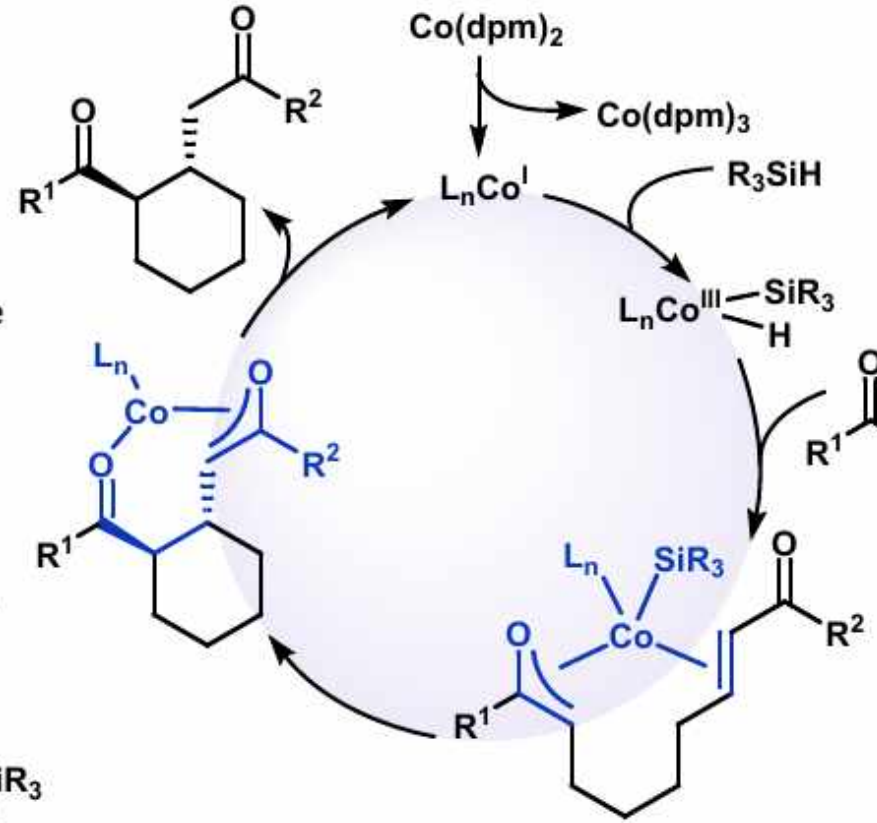
■ Some mechanistic proposed in the past

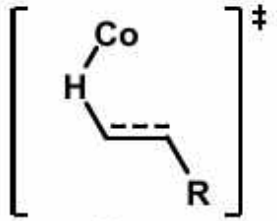


Sterically disfavored

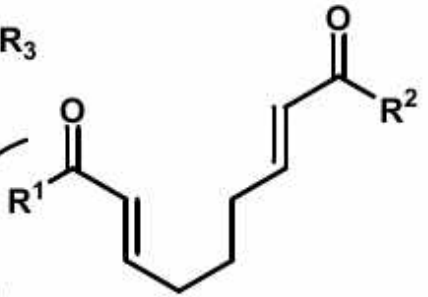
Proposed catalytic cycle for the reductive cycloaddition to an aldehyde.

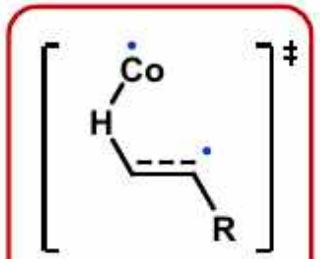




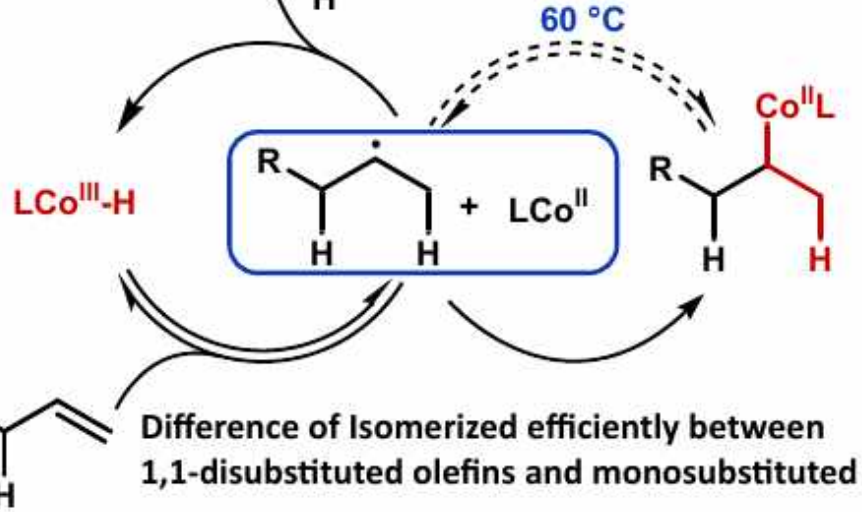


Very high pKa





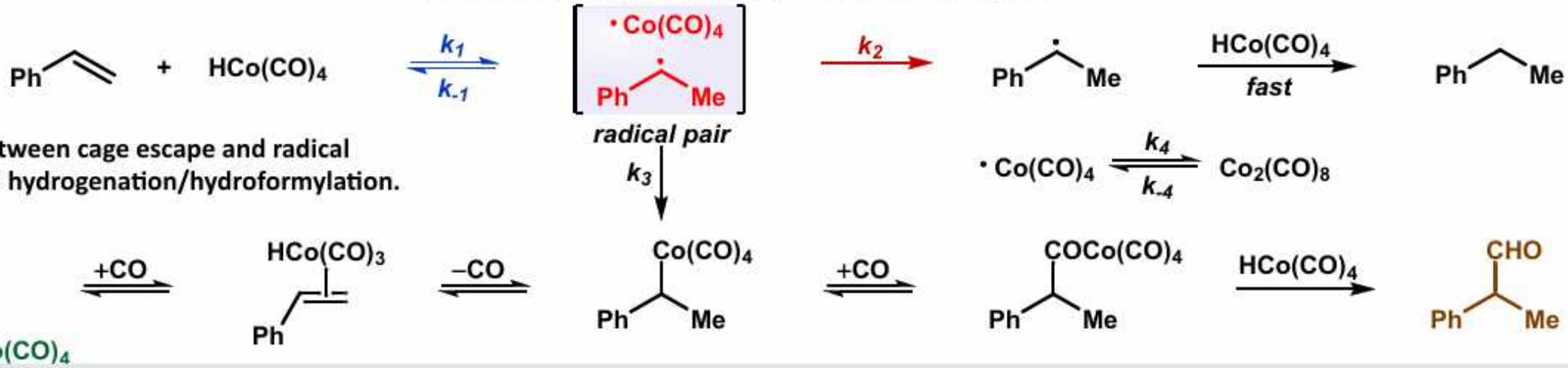
Markovnikov preferred



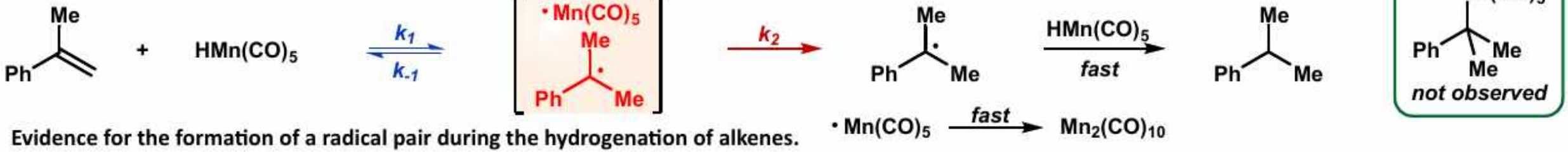
Difference of Isomerized efficiently between 1,1-disubstituted olefins and monosubstituted olefins

Mechanistic Overview and a more Realistic Situation

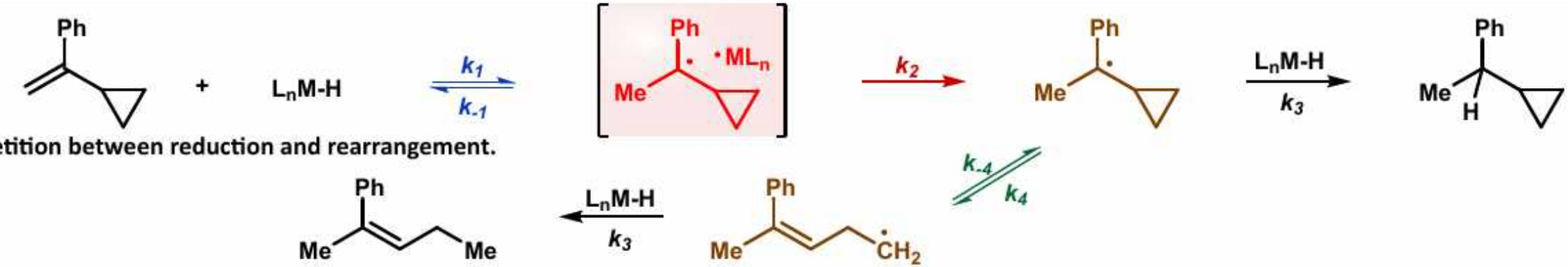
■ Some mechanistic proposed in the past



Competition between cage escape and radical collapse toward hydrogenation/hydroformylation.



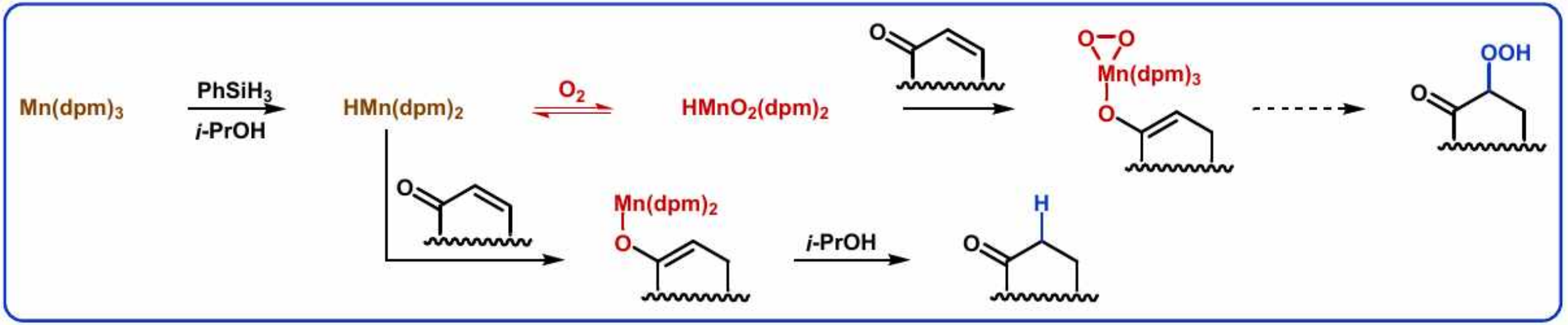
Evidence for the formation of a radical pair during the hydrogenation of alkenes.



Competition between reduction and rearrangement.

Mechanistic Overview and a more Realistic Situation

■ Some mechanistic proposed in the past



■ Different systems of two kinds of ligands

WF Systems

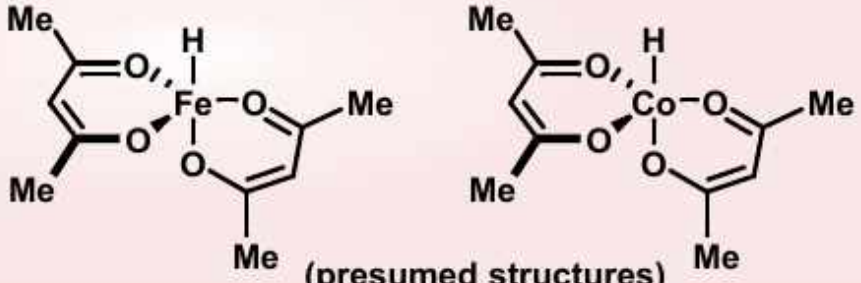
Not isolable

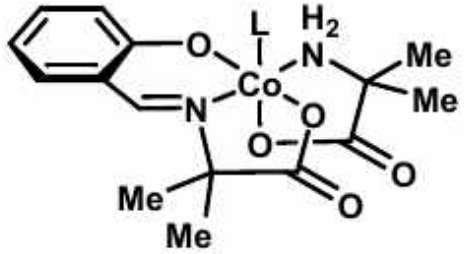
Weak Field (WF)ligands

Intermediate- or high-spin configurations

3+ oxidation states

Weaker M-H bonds





Chemical reviews **2016**, *116*, 8912.

SF Systems

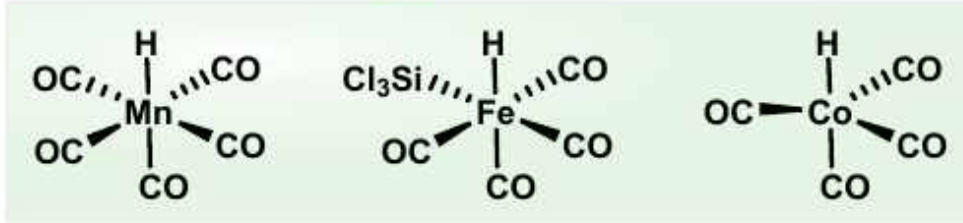
Isolable

Strong Field (SF)ligands

Low-spin electronic configurations

1+ and 2+ oxidation states

Stronger M-H bonds

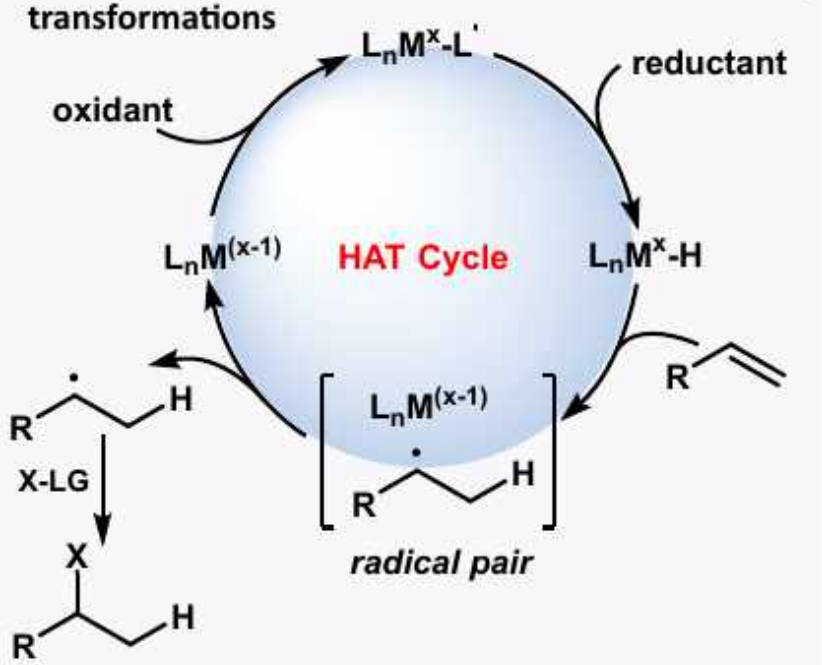


Chemical reviews **2016**, *116*, 8912.

Mechanistic Overview and a more Realistic Situation

■ Begin with formation of metal hydride

An MH-HAT mechanism to net redox-neutral catalytic transformations



Chemical reviews **2016**, *116*, 8912.

L_nM-Z + $H-SiR_3$ ⟶ L_nM-H + $Z-SiR_3$

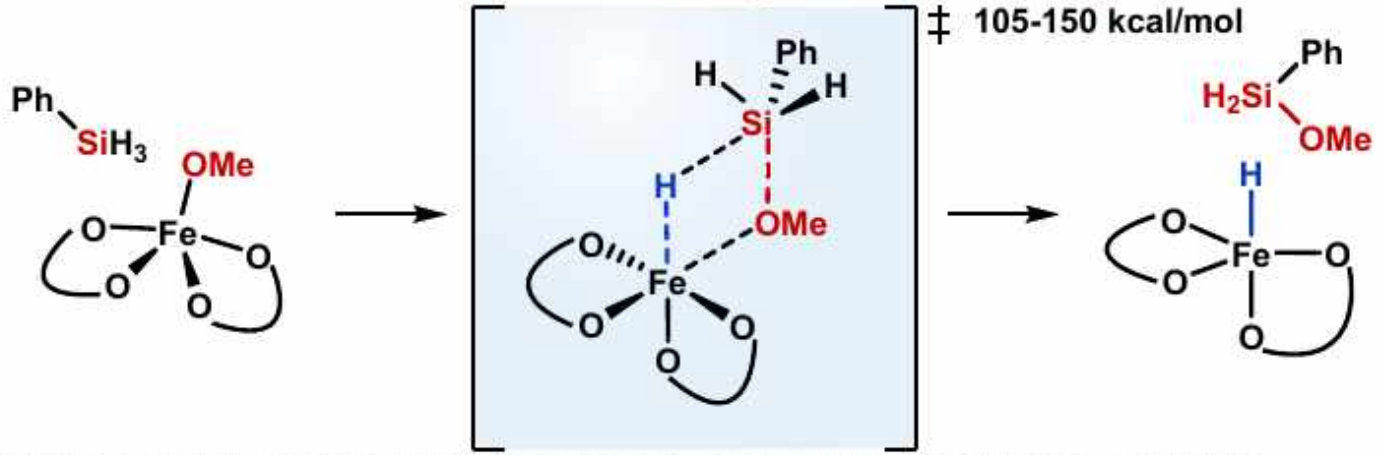
Z = OR, F, OOR, OTf

hydride precursor 70-95 kcal/mol

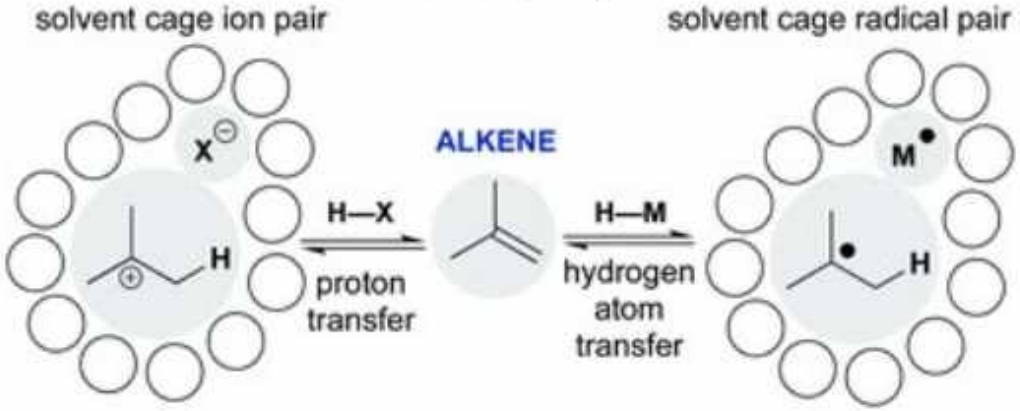
weak bond provides high reactivity

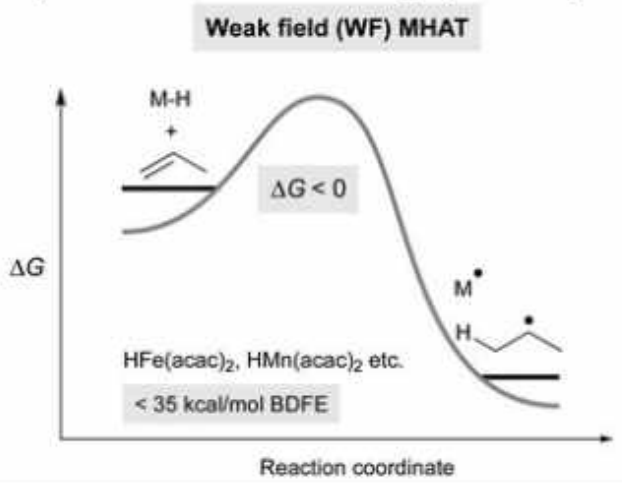
strong bond provides driving force

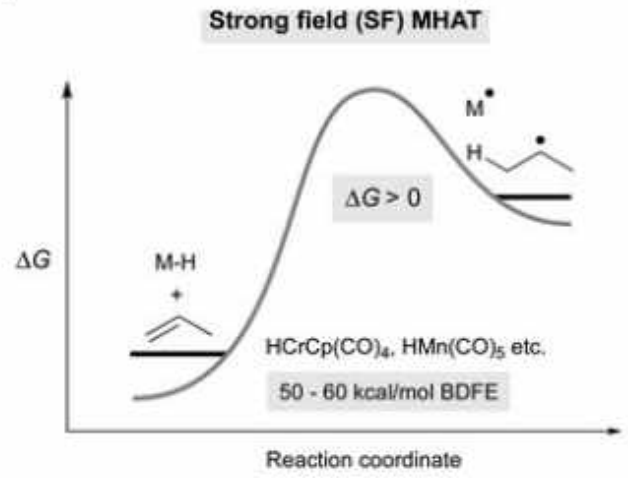
‡ 105-150 kcal/mol



Metal hydride formation through ligand exchange could occur through associative, dissociative, or concerted interchange pathways. Only propose the concerted interchange here.



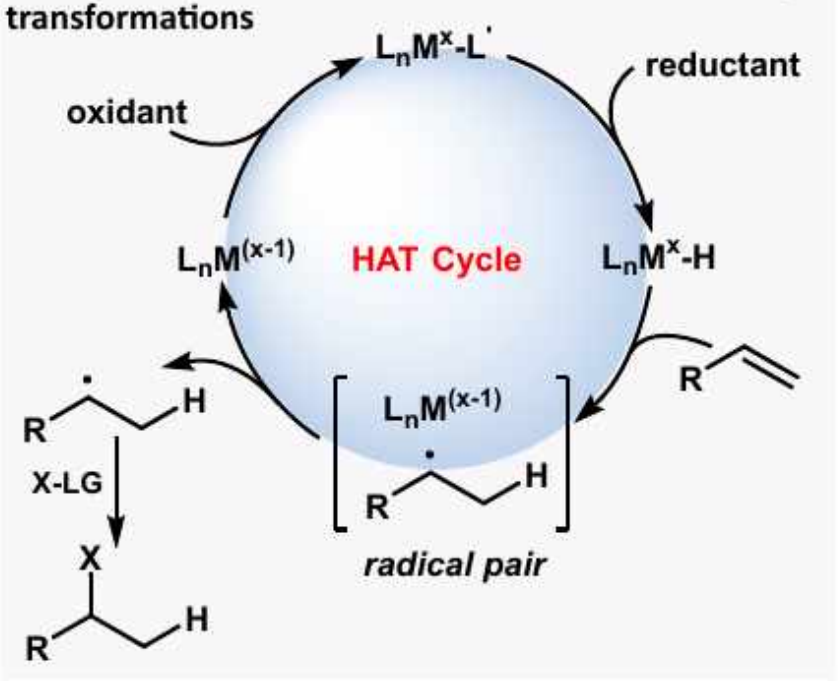




Mechanistic Overview and a more Realistic Situation

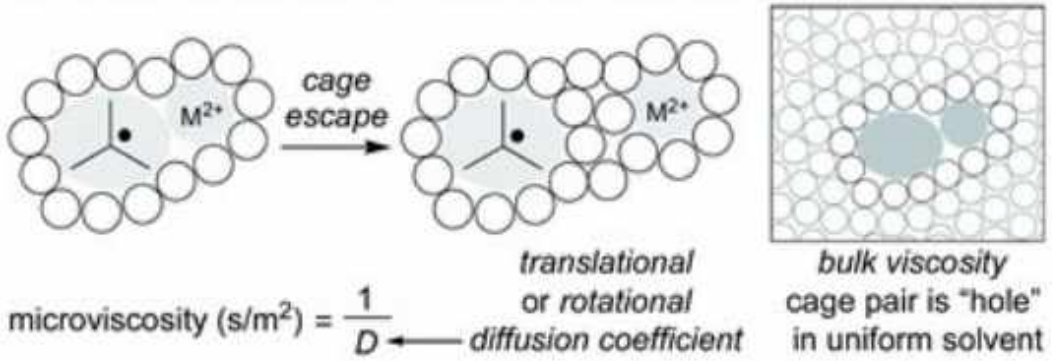
■ Reactions from the radical pair

An MH-HAT mechanism to net redox-neutral catalytic transformations

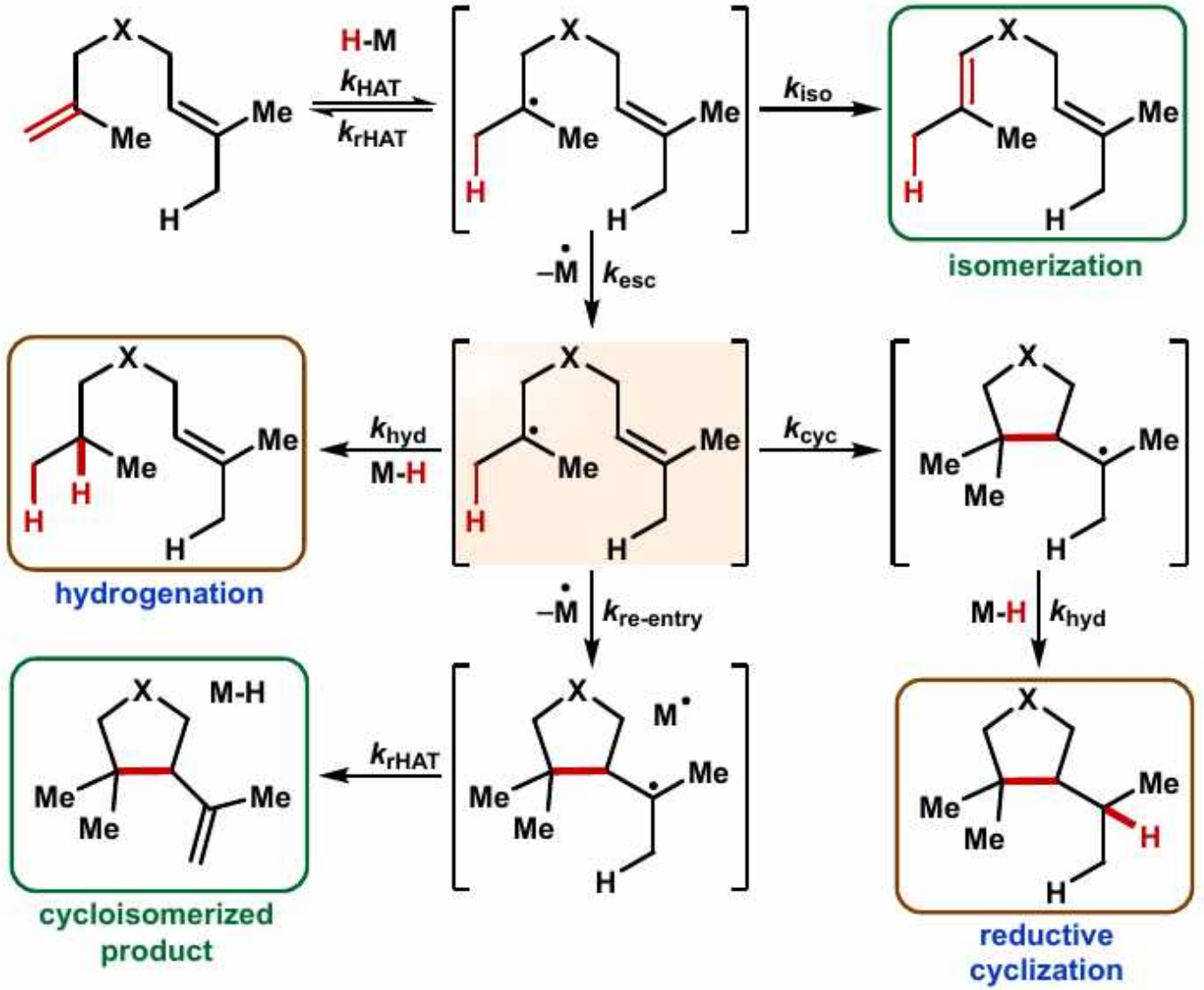


Chemical reviews **2016**, *116*, 8912.

microviscosity better predicts cage effects than bulk solvent parameters



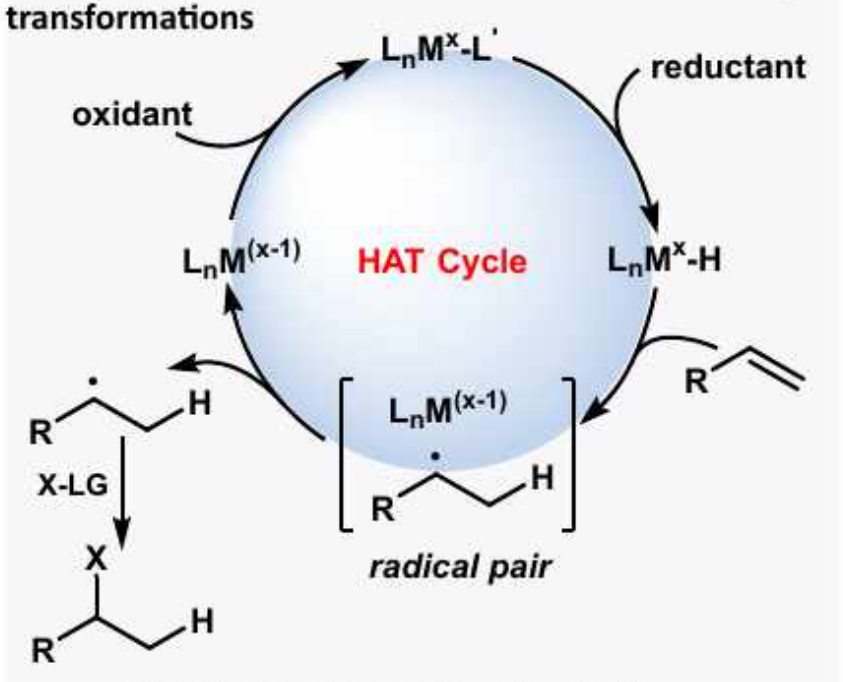
Competing pathways: hydrogenation, isomerization, and cyclization



Possibility of asymmetric MH-HAT reactions from the mechanism

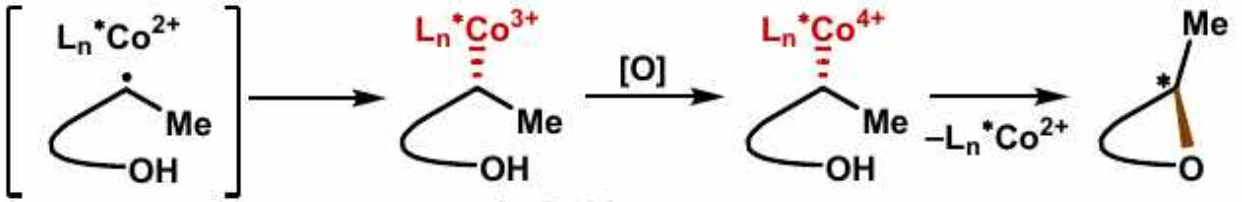
■ Stereochemical retention from putative radical-pair intermediates

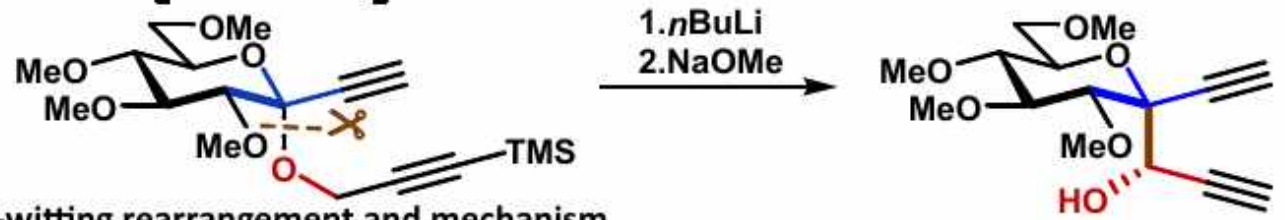
An MH-HAT mechanism to net redox-neutral catalytic transformations



Chemical reviews **2016**, *116*, 8912.

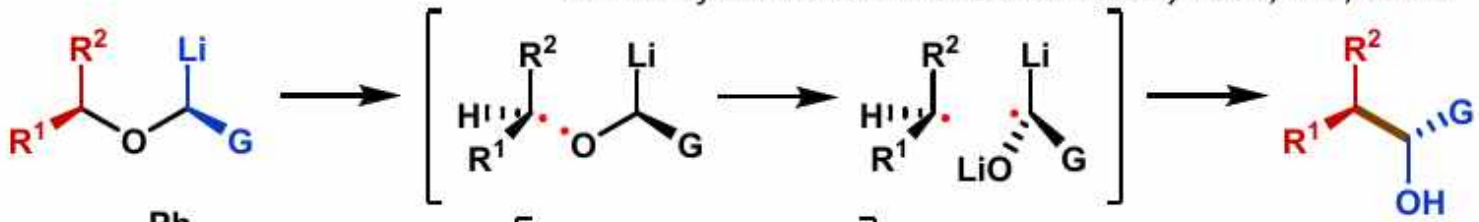
Primary geminate cage collapse to organocobalt(III)

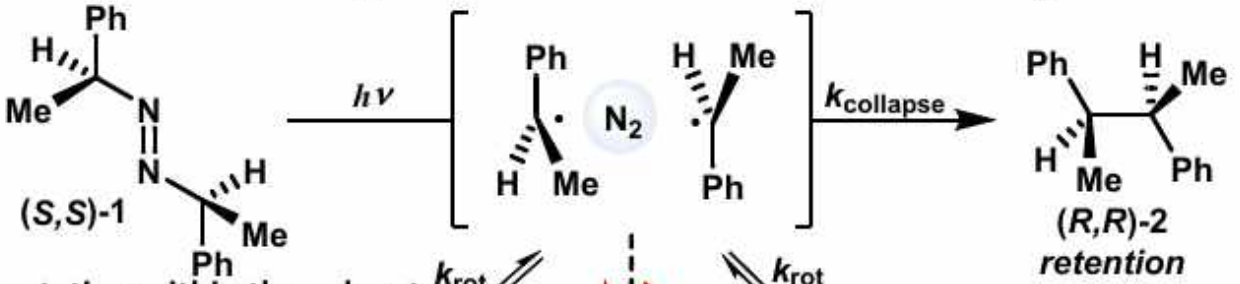




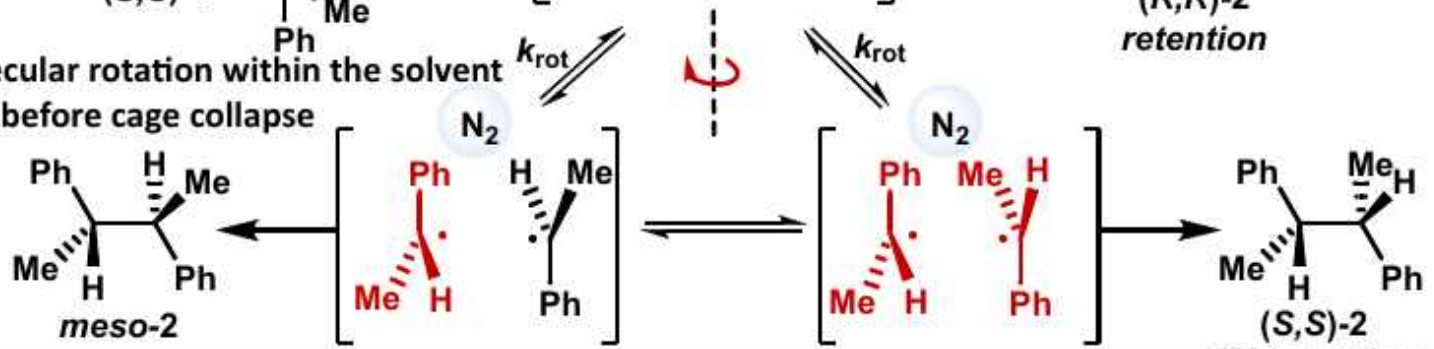
1,2-witting rearrangement and mechanism

Journal of the American Chemical Society **1996**, *118*, 3317.

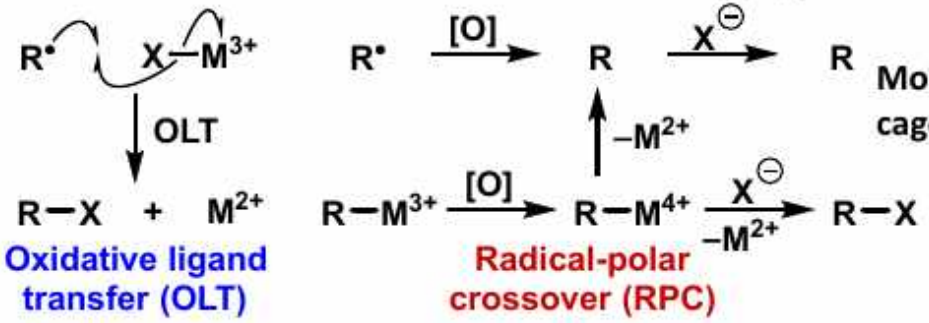




Molecular rotation within the solvent cage before cage collapse



Mechanisms with metal-mediated radical trapping

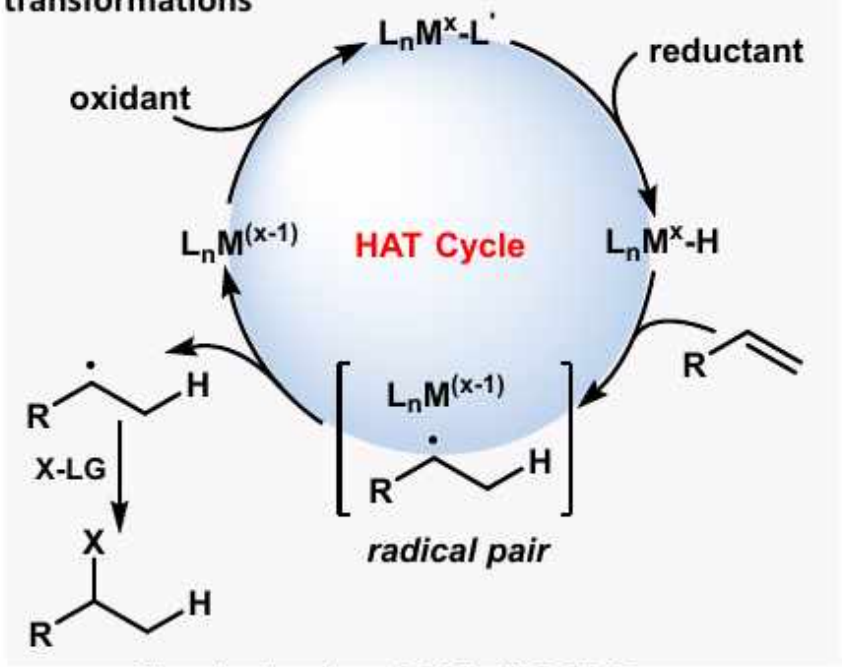


Shevick, S. L.; Wilson, C. V.; Kotesova, S.; Kim, D.; Holland, P. L.; Shenvi, R. A. *Chemical Science* **2020**, *11*, 12401.

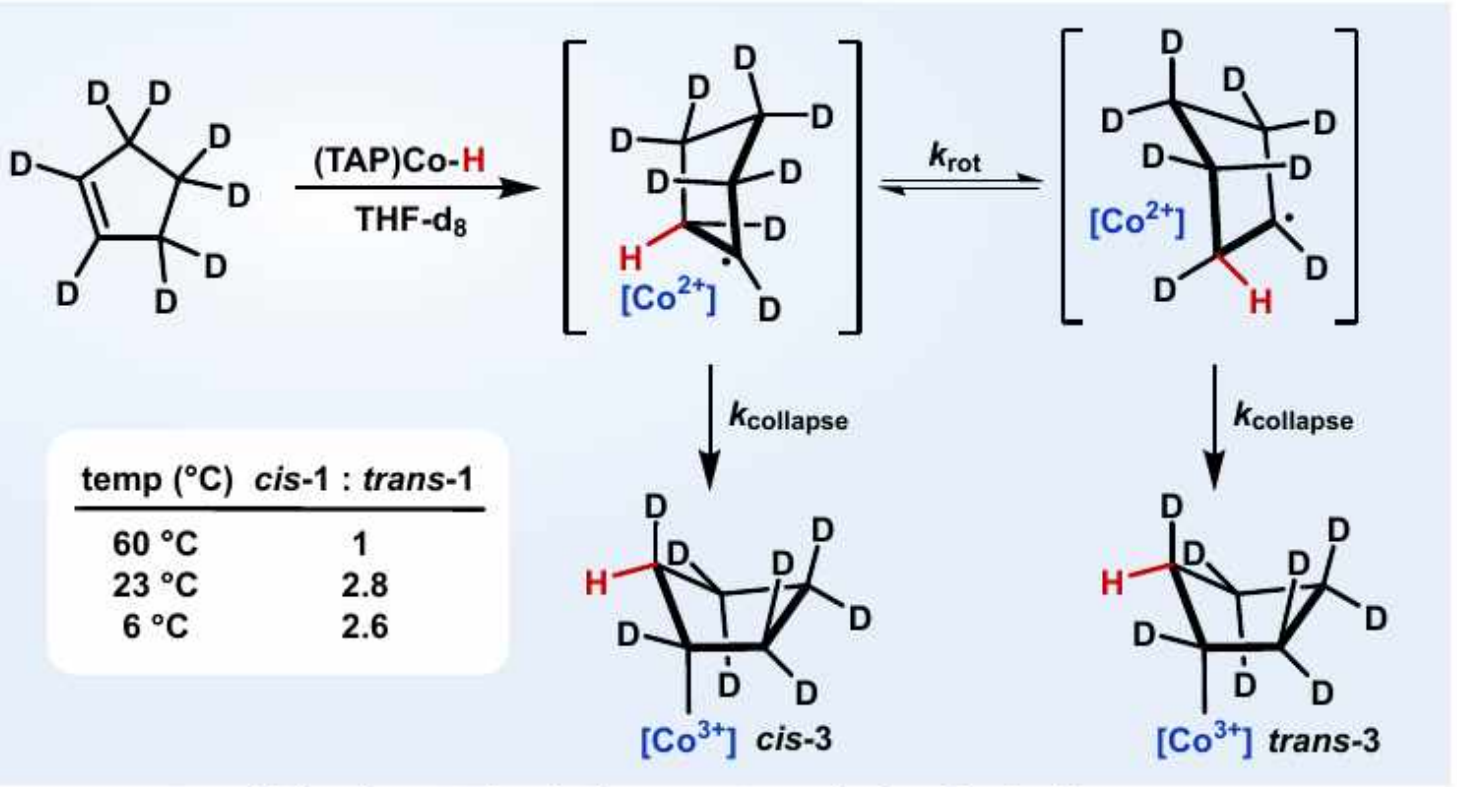
Possibility of asymmetric MH-HAT reactions from the mechanism

■ Stereochemical retention from putative radical-pair intermediates

An MH-HAT mechanism to net redox-neutral catalytic transformations

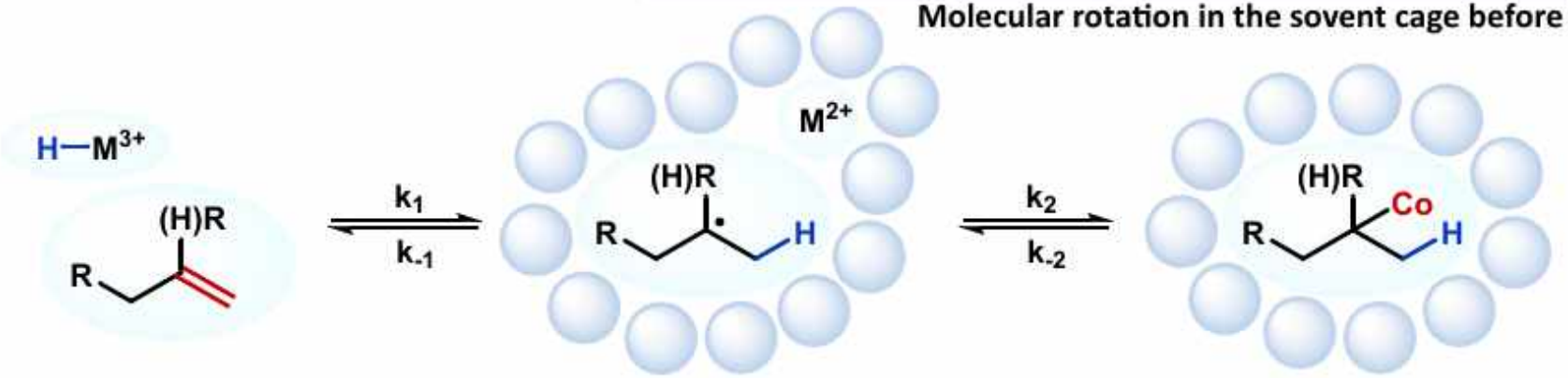


Chemical reviews **2016**, *116*, 8912.



temp (°C)	*cis*-1 : *trans*-1
60 °C	1
23 °C	2.8
6 °C	2.6

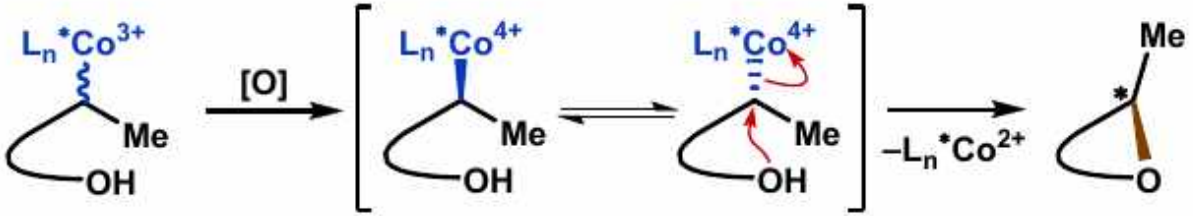
Molecular rotation in the sovent cage before Co-C collapse



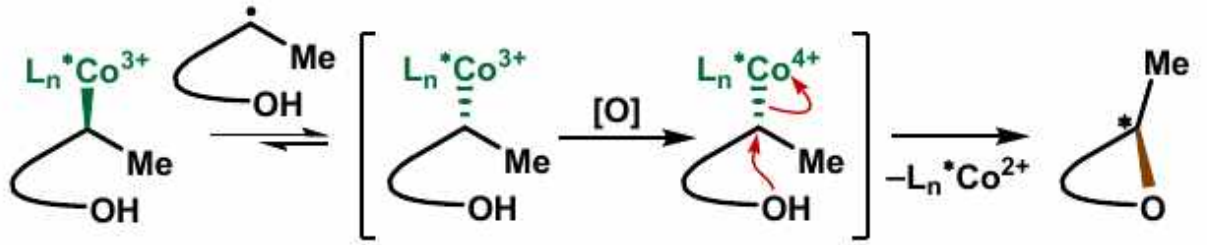
Possibility of asymmetric MH-HAT reactions from the mechanism

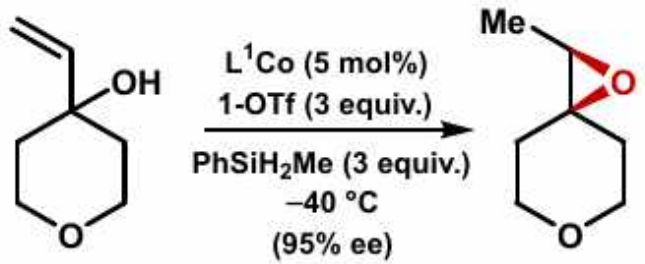
■ Some mechanistic proposed in the past

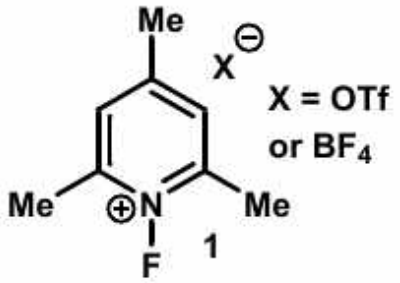
Dynamic Kinetic Resolution of oxidized organocobalt(IV)

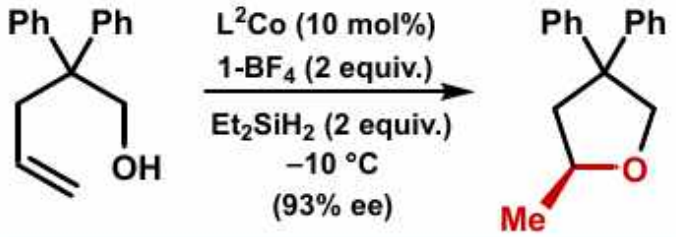


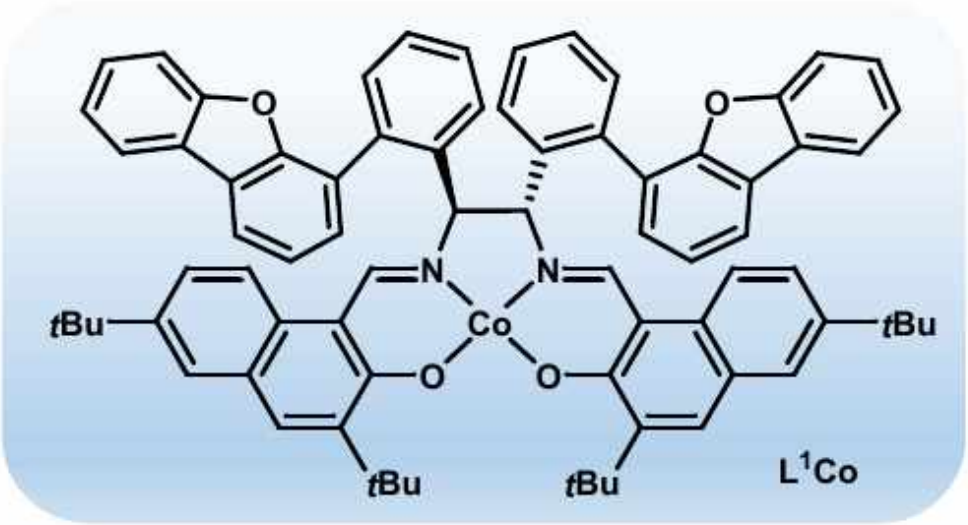
Radical Chain leads to enantioenriched organocobalt(III)







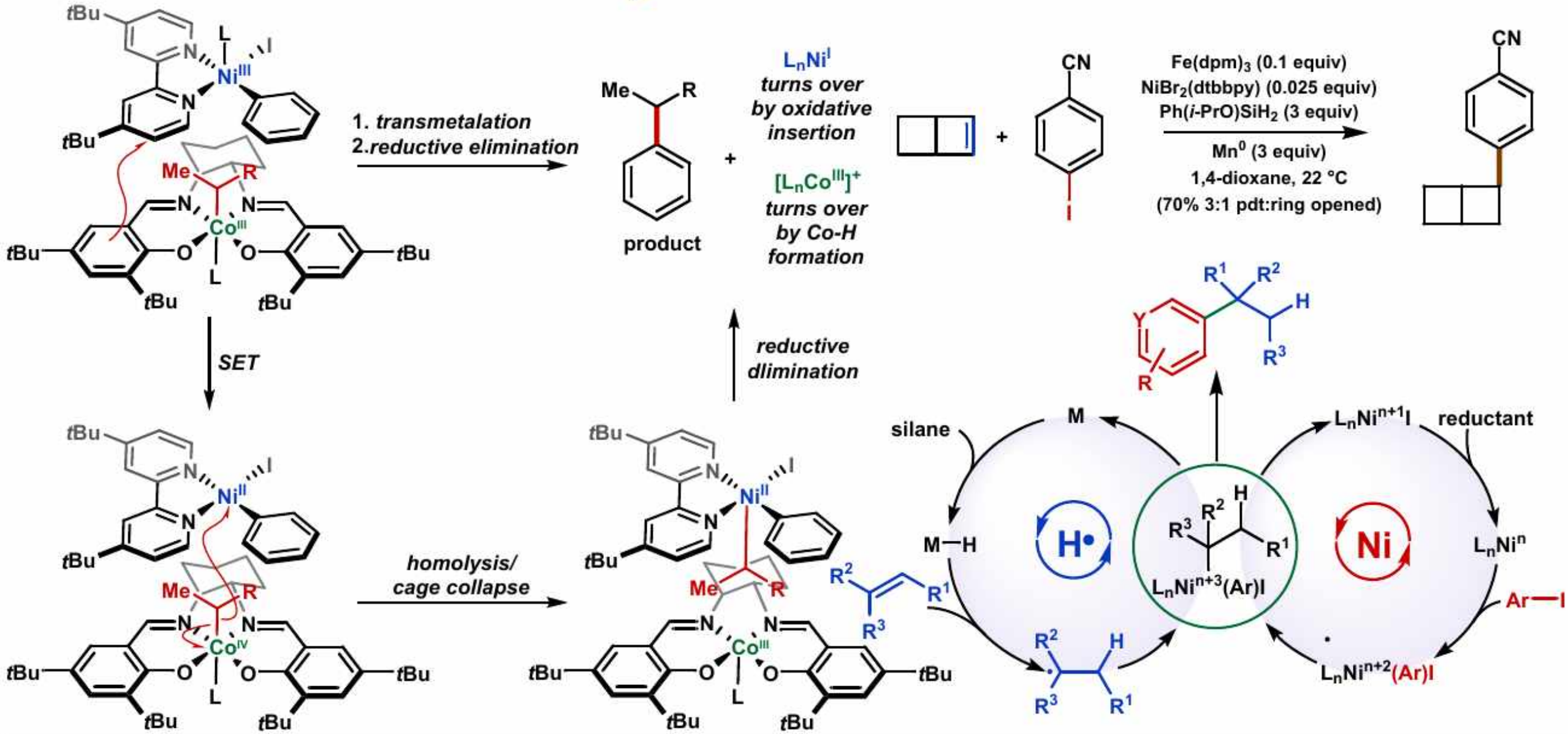






Shevick, S. L.; Wilson, C. V.; Kotesova, S.; Kim, D.; Holland, P. L.; Shenvi, R. A. *Chemical Science* **2020**, *11*, 12401.

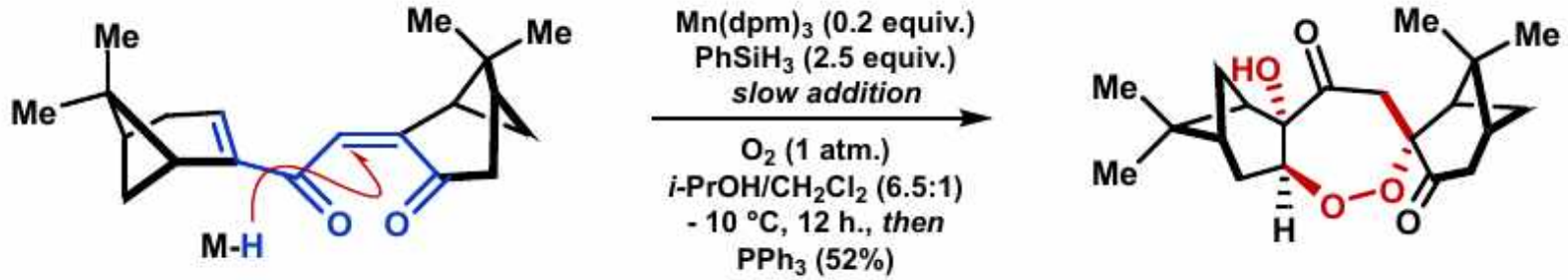
IRON-BASED MH-HAT/NI Chemistry

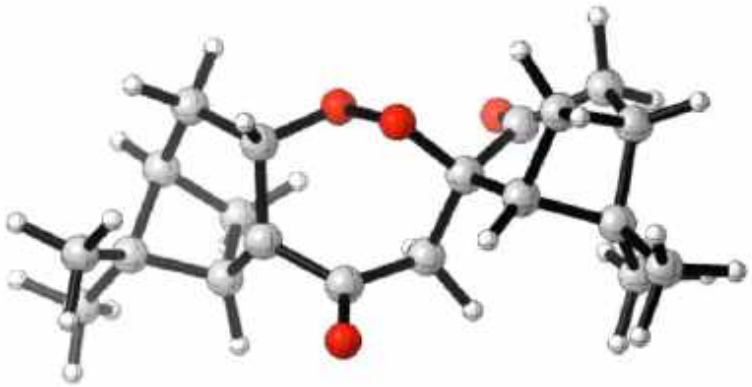


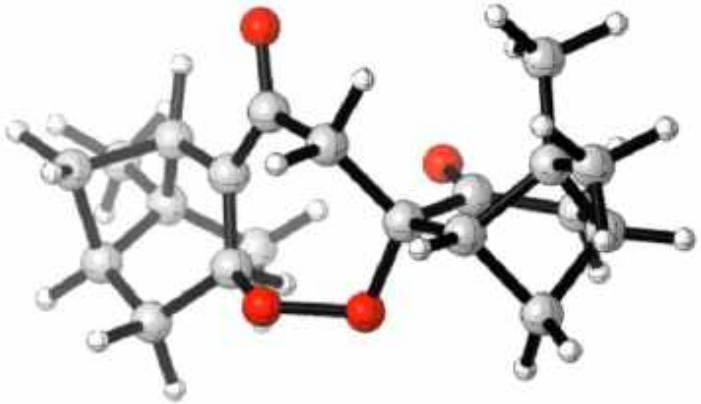
Green, S. A.; Crossley, S. W.; Matos, J. L.; Vásquez-Céspedes, S.; Shevick, S. L.; Shenvi, R. A. *Accounts of chemical research* **2018**, *51*, 2628.

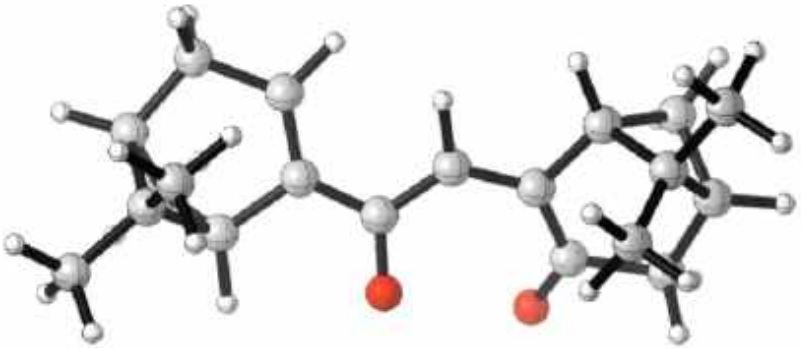
Simple MH-HAT Reaction in nature product and our exams

■ Maimone's synthesis of (+)-cardamom peroxide via a magic manganese-catalyzed peroxidation



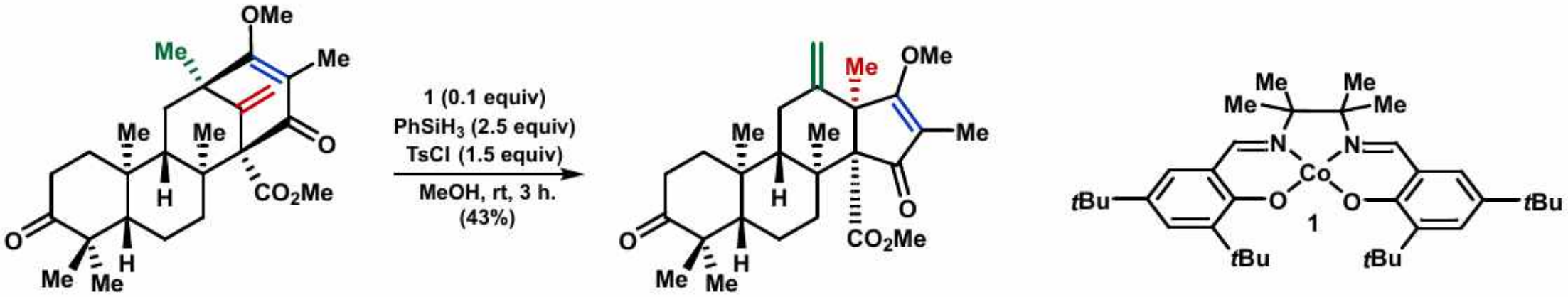


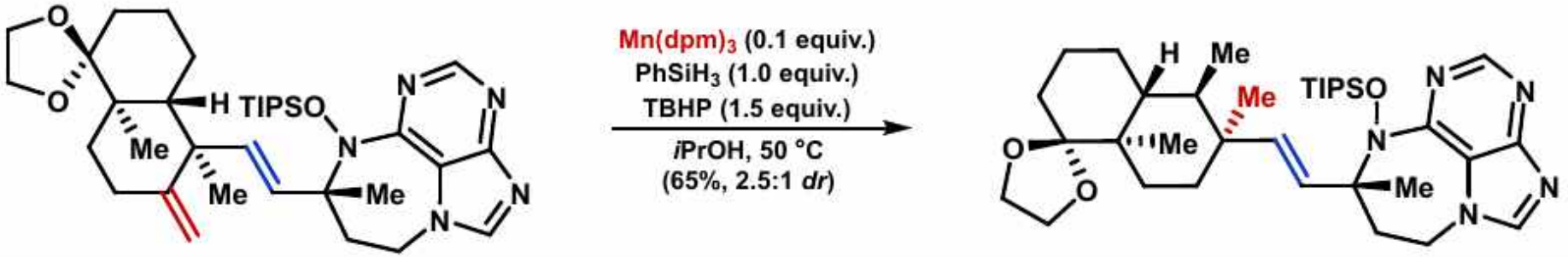




Simple MH-HAT Reaction in nature product and our exams

■ The classic homoallyl radical rearrangement occurred in MH-HAT catalysis

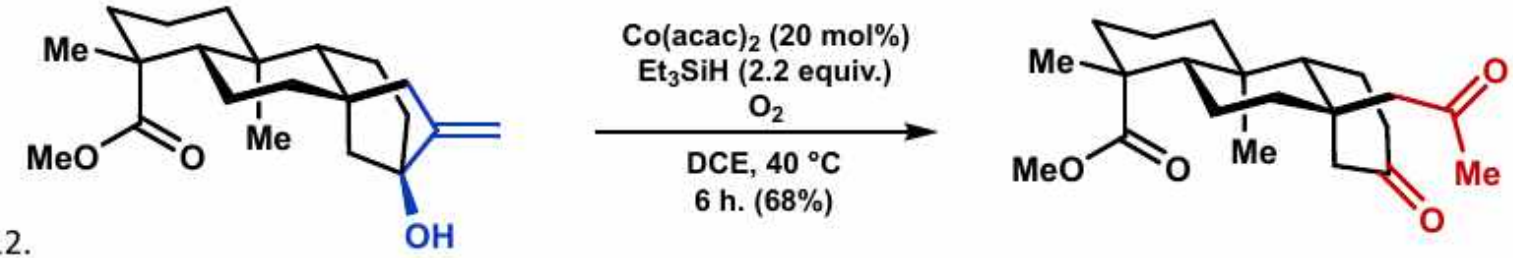




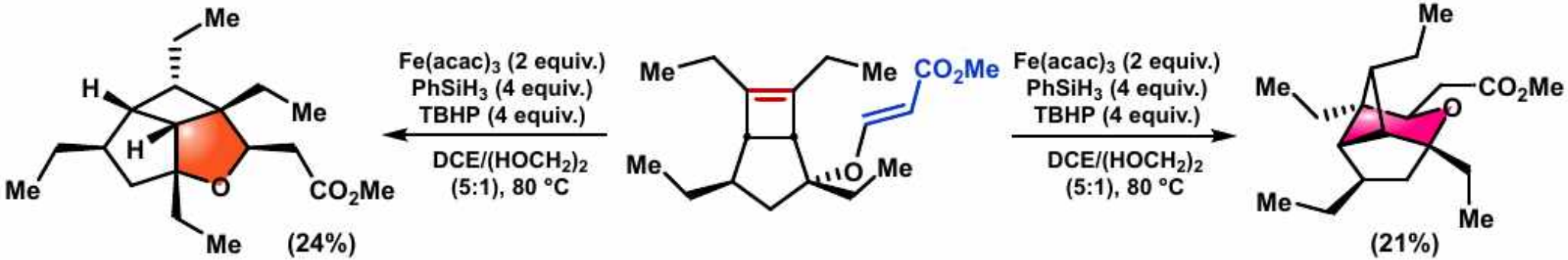
Green, S. A.; Crossley, S. W.; Matos, J. L.; Vásquez-Céspedes, S.; Shevick, S. L.; Shenvi, R. A. *Accounts of chemical research* **2018**, *51*, 2628.

Simple MH-HAT Reaction in nature product and our exams

- Carbon skeleton rearrangement and stereoselective construction of carbon skeleton



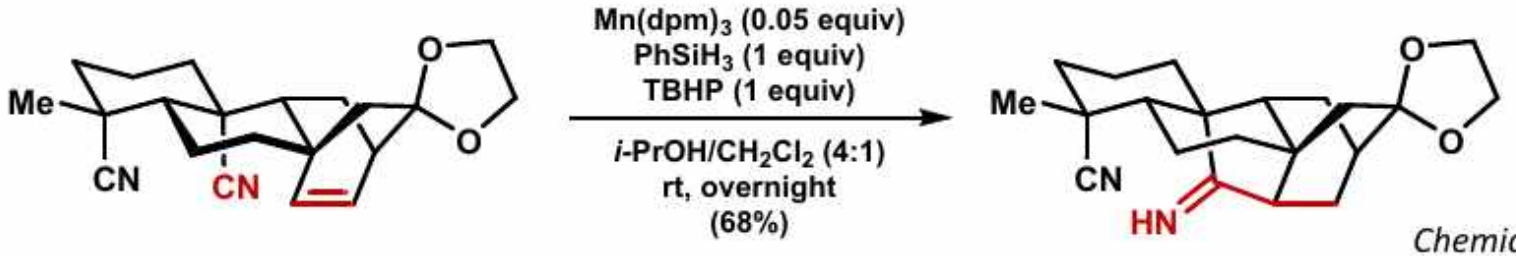
Chemical reviews **2016**, *116*, 8912.



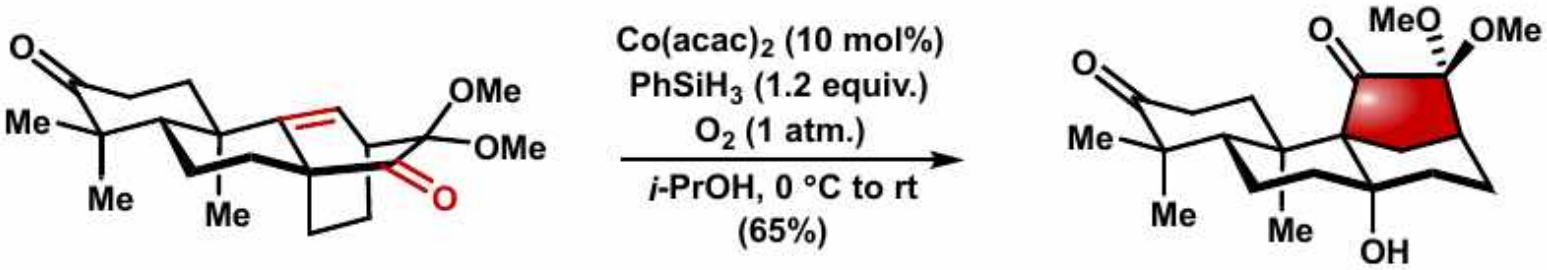
Chemical reviews **2016**, *116*, 8912.

Simple MH-HAT Reaction in nature product and our exams

- Carbon skeleton rearrangement and stereoselective construction of carbon skeleton



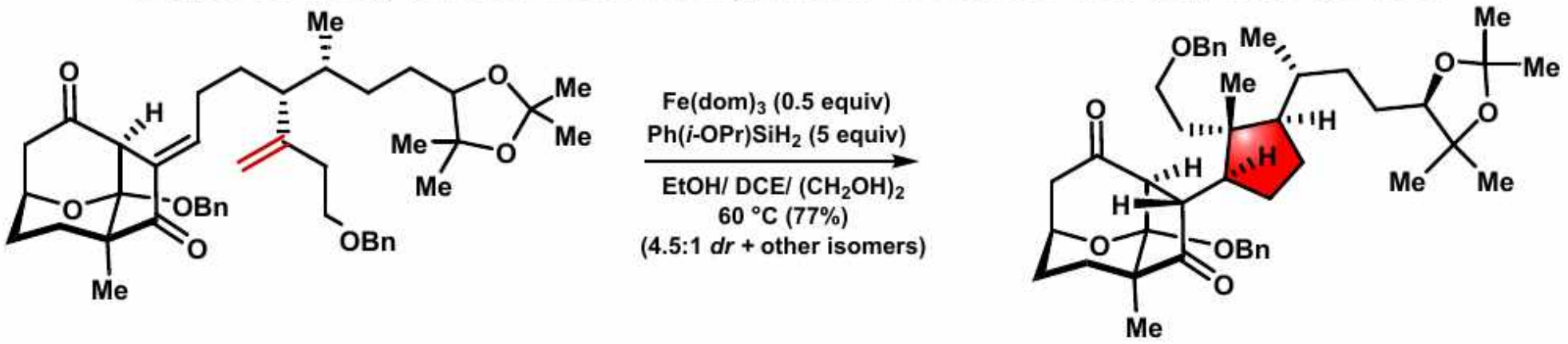
Chemical reviews **2016**, *116*, 8912.

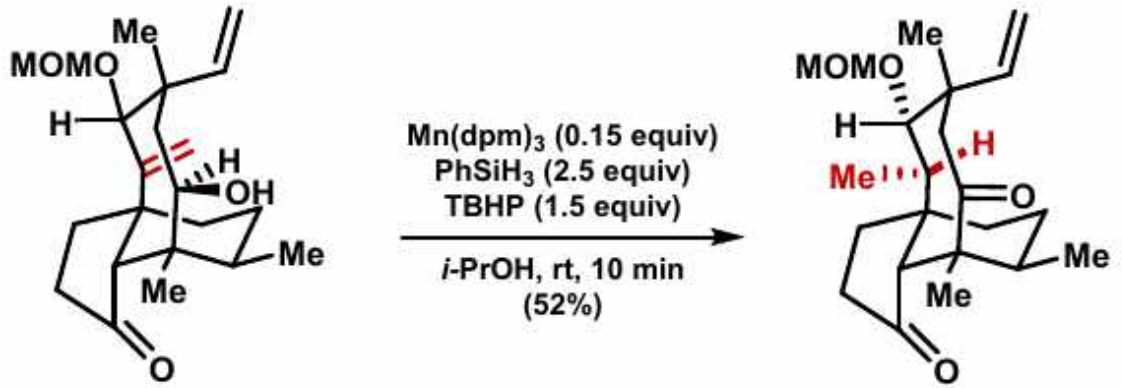


Green, S. A.; Crossley, S. W.; Matos, J. L.; Vásquez-Céspedes, S.; Shevick, S. L.; Shenvi, R. A.
Accounts of chemical research **2018**, *51*, 2628.

Simple MH-HAT Reaction in nature product and our exams

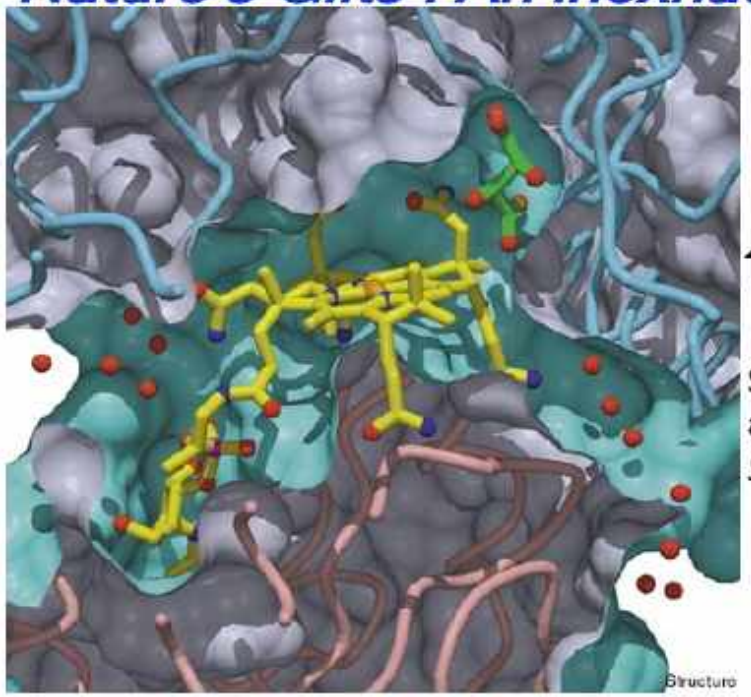
■ Regioselectivity depend on thermodynamics or kinetics and a special 1,5-HAT



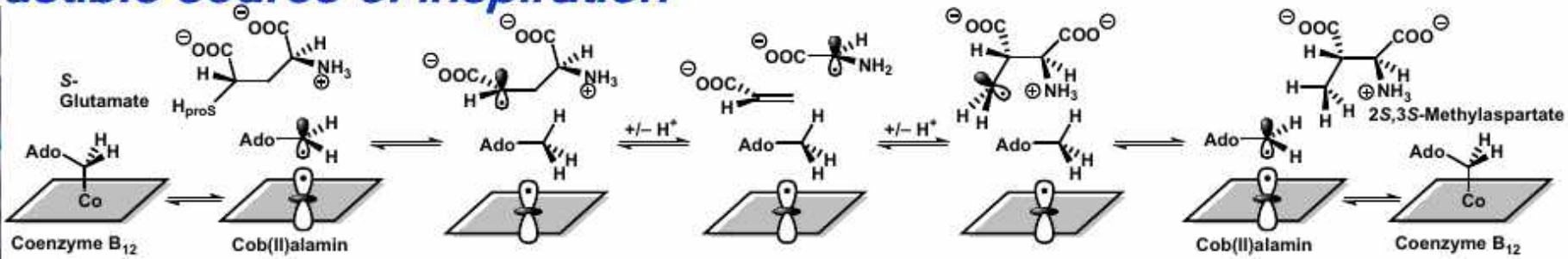


Green, S. A.; Crossley, S. W.; Matos, J. L.; Vásquez-Céspedes, S.; Shevick, S. L.; Shenvi, R. A. *Accounts of chemical research* **2018**, *51*, 2628.

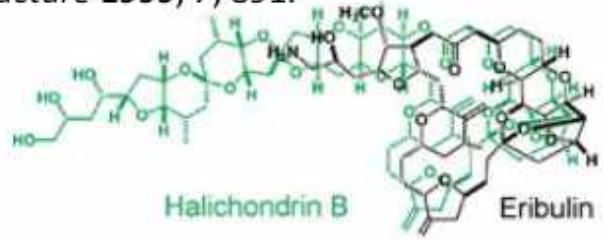
Nature's Gifts : An inexhaustible source of inspiration



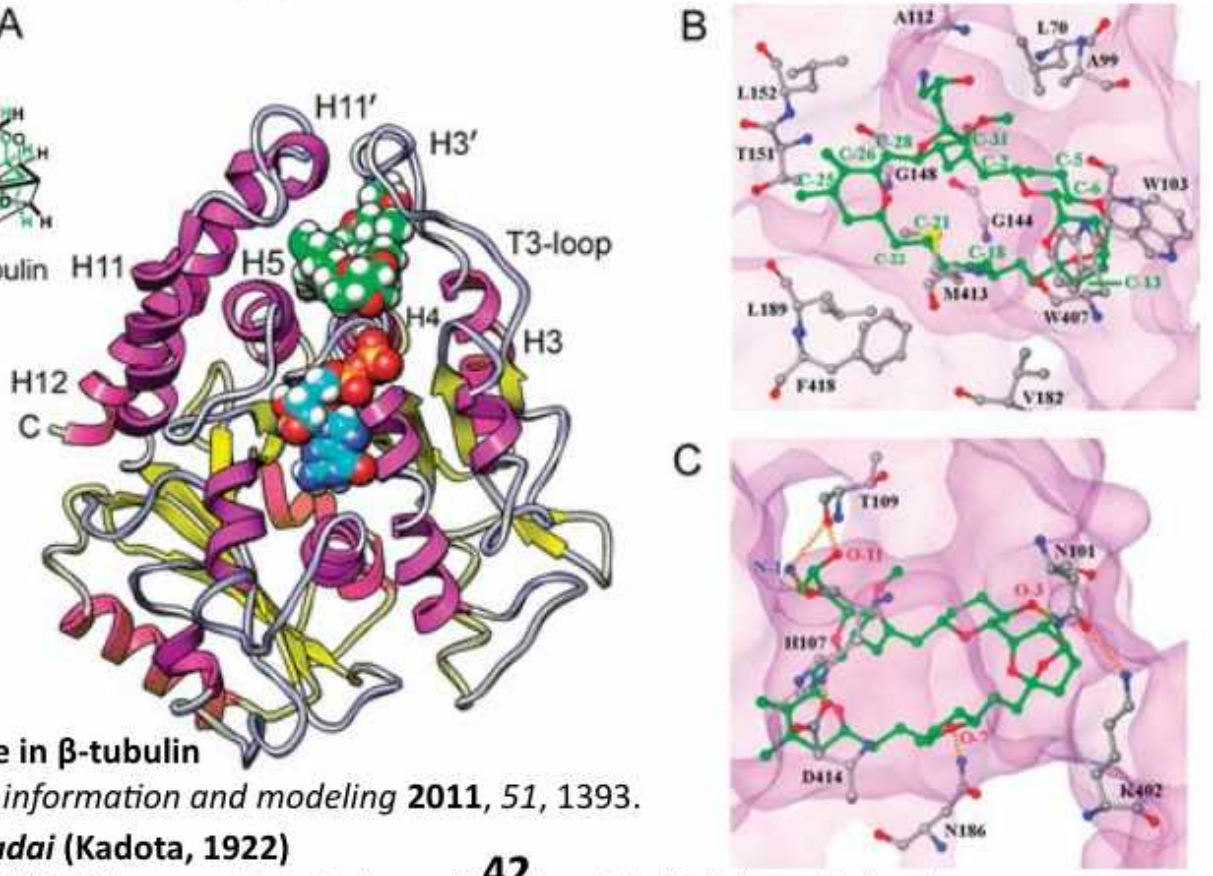
Surface representation of the B12- and tartrate-binding region
Structure **1999**, *7*, 891.



Chemical structures of the B_{12} cofactor and proposed reaction intermediates.
Structure **1999**, *7*, 891.



Halichondrin B and Eribulin
Journal of chemical information and modeling **2011**, *51*, 1393.



Eribulin binding site in β-tubulin
Journal of chemical information and modeling **2011**, *51*, 1393.



***Halichondria* (*Halichondria*) *okadai* (Kadota, 1922)**
https://tonysharks.com/Tree_of_life/Eukaryote/Opisthokonta/Halichondria/Halichondria.html